Microeconomics

Lecture 3: Classical Demand Theory

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Outline

- Basic properties of preference relation
- Utility function
- Utility maximization
- Expenditure minimization
- Duality
- Indirect Utility
- Integrability
- Welfare evaluation
- Strong axiom of revealed preferences

Preferences

Rational preferences: completeness, transitivity.

We introduce more properties:

- desirability: monotonicity, local nonsatiation
- convexity
- homotheticity, quasilinearity, etc.

Desirability assumption says the more the better.

Convexity assumption says we prefer the middle to the extremes (moderation of all things)

Monotonicity

Definition (monotonicity)

The preference relation $\succsim$ on X is monotone if $x \in X$ and $y >> x$ implies $y \succ x$ and is strongly monotone if $y \geq x$ and $y \neq x$ imply $y \succ x$.

This assume all commodities are goods not bads. Is it reasonable?

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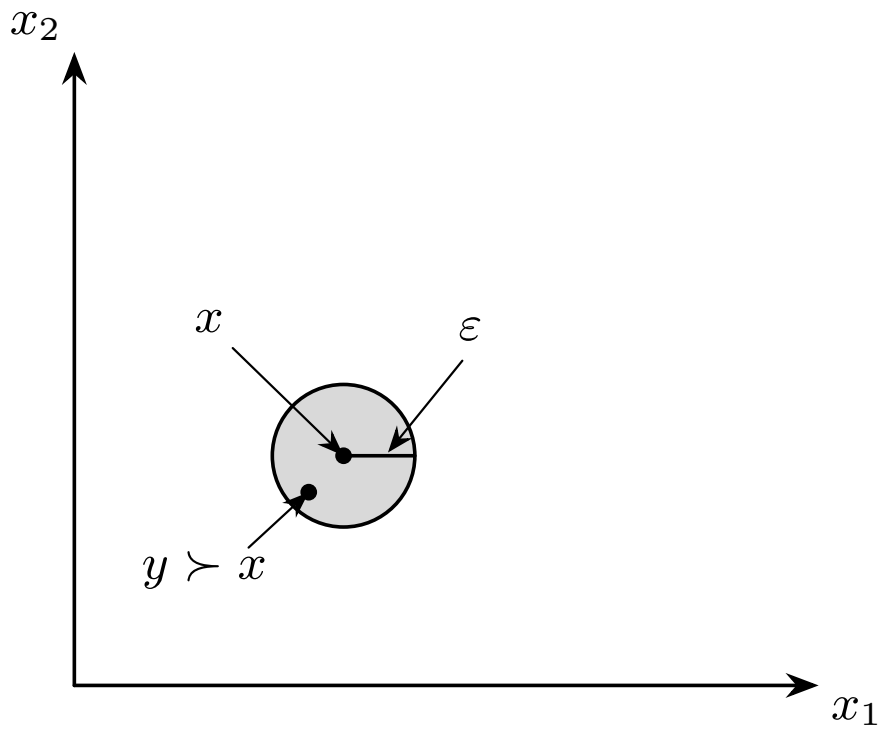
- pollution vs. cleanness, garbage vs. absence of garbage
- do we always prefer more sweetness and fat?
- do you prefer more sodium bi-sulphate?
- consumption or available for consumption?
- free disposal?

Local nonsatiation

A less restrictive assumption:

Definition (local nonsatiation)

$\succsim$ on X is locally nonsatiated if for every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ such that $||y - x|| \leq \varepsilon$ and $y \succ x$.

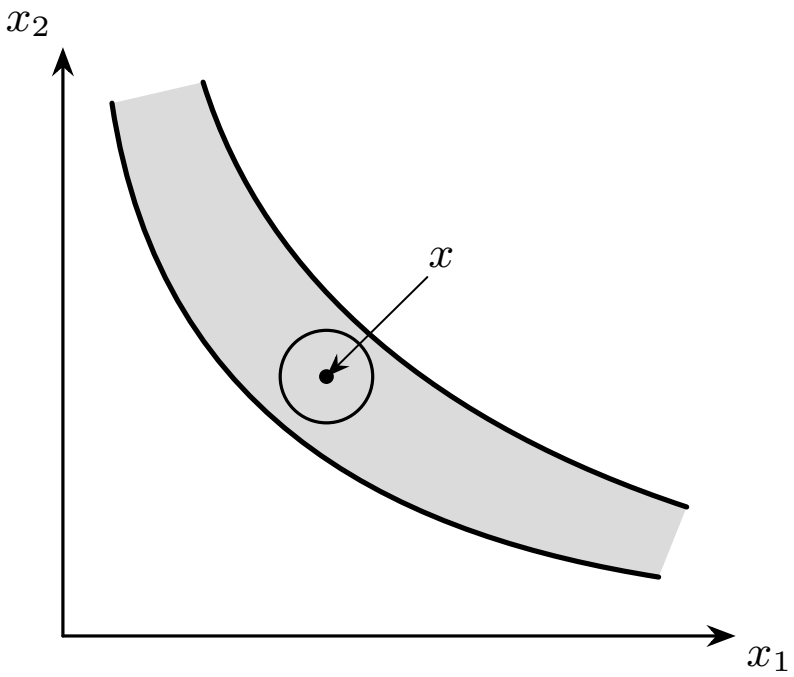


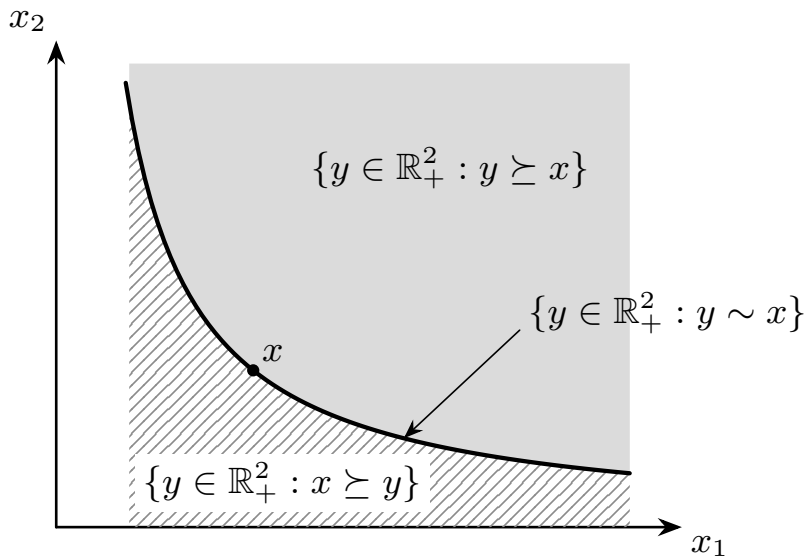
Local nonsatiation

- Local nonsatiation means there is no peak in preferences.
- It is implied by monotonicity of preferences but not vice-versa.
- There is no requirement that the preferred bundle y contain more of any good.
- Some goods can be "bads" and preferences can be non-monotone.
- If $x \in \mathbb{R}^L_+$, it rules out the extreme case where all goods are "bads", since the point $x = 0$ would then be a *bliss* point.
- Local nonsatiation can only occur if the consumption set is either unbounded (open) or on sections of a bounded consumption set sufficiently far away from the ends.
- Near the ends of a bounded set, there would necessarily be a bliss point where local nonsatiation doesn't hold.

Local nonsatiation

Local nonsatiation rules out "thick" indifference set.

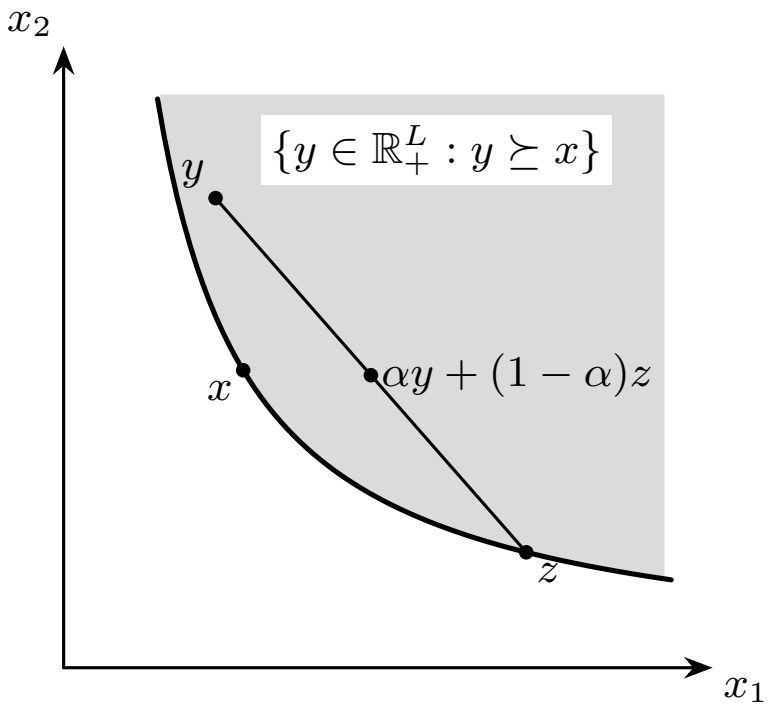


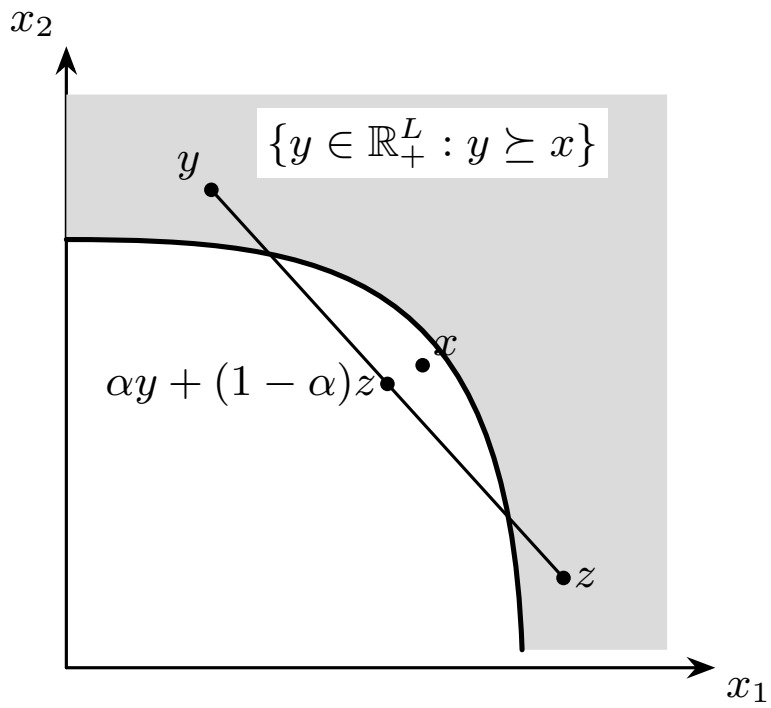


Convexity

Definition (convexity)

$\succsim$ is convex if $y \succsim x$ and $z \succsim x$, then $\forall \alpha \in [0,1]$, $\alpha y + (1-\alpha)z \succsim x$.



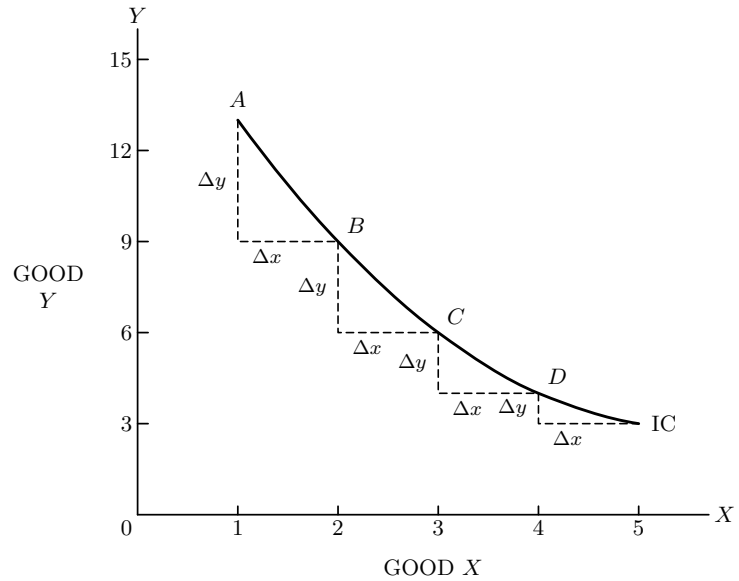


Convexity

The **marginal rate of substitution** of X for Y measures the number of units of Y that must be scarified for unit of X gained so as to maintain a constant level of satisfaction.

Convexity is equivalent to **diminishing** marginal rate of substitution.

Diminishing MRS: as the consumer has more and more units of good X, he is prepared to forego less and less of good Y.



Convexity

Convexity also reflects the inclination for diversification.

If $y \succsim x$, then on the direct path from x to y, we never worse off.

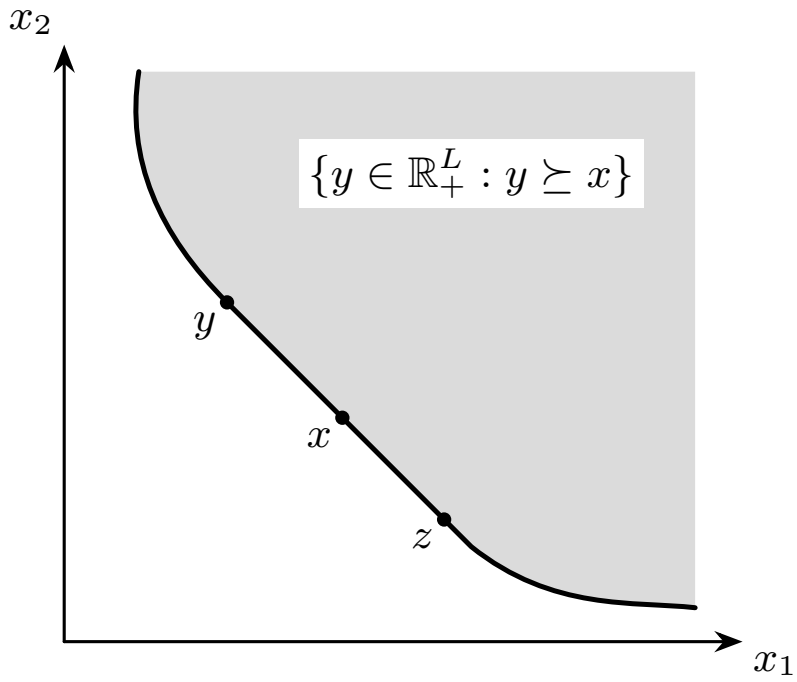
We'll see that convexity of preferences is equivalent to the "quasi-concavity" of the utility function.

Strict convexity

Definition (strict convexity)

$\succsim$ is strictly convex if $y \succsim x$, $z \succsim x$, and $y \neq z$ implies $\forall \alpha \in [0,1]$, $\alpha y + (1-\alpha)z \succ x$.

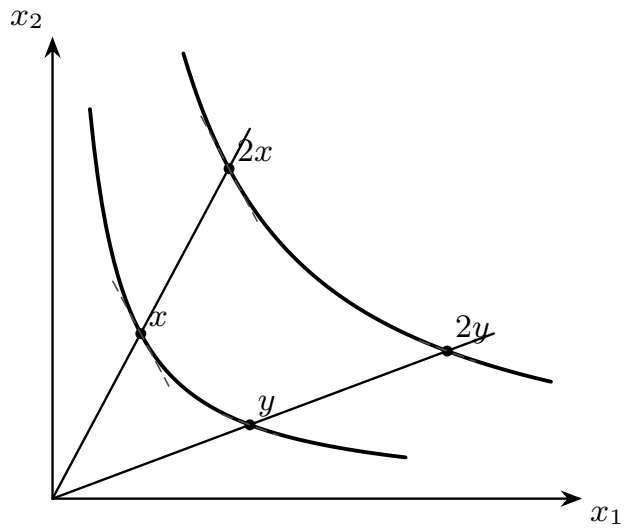
A convex but not strictly convex preference relation



Homotheticity

Definition (homotheticity)

A monotone $\succsim$ on $\mathbb{R}^L_+$ is homothetic if $y \sim x$ implies $\forall \alpha \in [0,1]$, $\alpha y \sim \alpha x$.



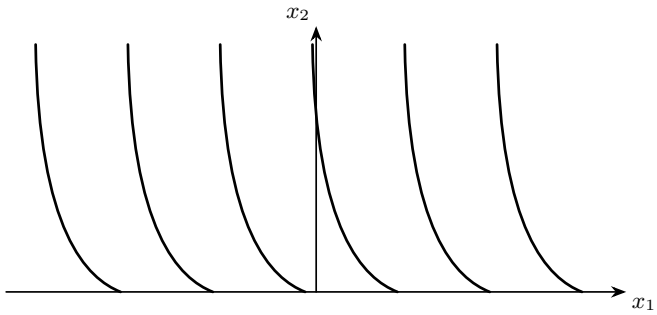
- Homothetic preferences have utility function homogeneous of degree one.
- Equivalent to constant return to scale
- Note that $\succsim$ is monotone
- The indirect utility function is a linear function of wealth: $v(p,w) = v(p)w$
- Ex: Linear, Leontief, Cobb–Douglas, CES utilities.

Quasilinearity

Definition (quasilinearity)

$\succsim$ on $X = (-\infty, \infty) \times \mathbb{R}^L_+$ is quasilinear w.r.t a numeraire commodity if $\forall \alpha > 0$,

- $y \sim x$ implies $y + \alpha e_1 \sim x + \alpha e_1$
- $x + \alpha e \succ x$

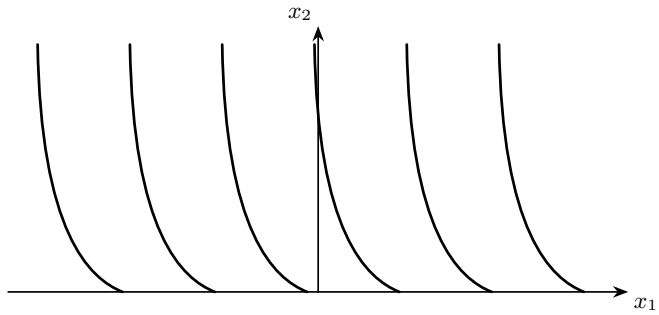


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utility is linear w.r.t. numeraire: $u(x_1, \ldots, x_n) = x_1 + \theta(x_2, \ldots, x_n)$.

- quasilinear utilities are, in general, not homothetic.
- there is no income effect on good 2 to L, all effect on good 1.
- often used to analyze goods constituting a small part of income.
- the numeraire can be thought of as "general consumption".

Preference and utility

Can all rational preference relations be represented by a utility function?

Lexcographic preference: complete, transitive, strongly monotone, and strictly convex, but with no utility representation.

It's impossible to find two bundles that are indifferent.

Each dimension has a ranking, but not possible to have all ranking in a real line.

What additional assumption one need to make to ensure the existence of utility function?

Continuity

Definition

continuity $\succsim$ on x is continuous, if for sequence $\{(x^n, y^n)\}_{n=1}^{\infty}$ with $x^n \succsim y^n$ and $x = \lim_{n\to\infty} x^n,\ y = \lim_{n\to\infty} y^n$, we have $x \succsim y$.

Lexcographic preference is not continuous: $x^n = (1/n, 0),\ y^n = (0, 1)$.

Continuity property can be restated by contour sets:
$\forall x \in X$ if the upper contour set $\{y \in X : y \succsim x\}$ and the lower contour set $\{y \in X : x \succsim y\}$ are both closed (i.e. include their boundaries).

Proposition

If the rational preference relation $\succsim$ on X is continuous, there is a continuous utility function $u(x)$ that represents $\succsim$.

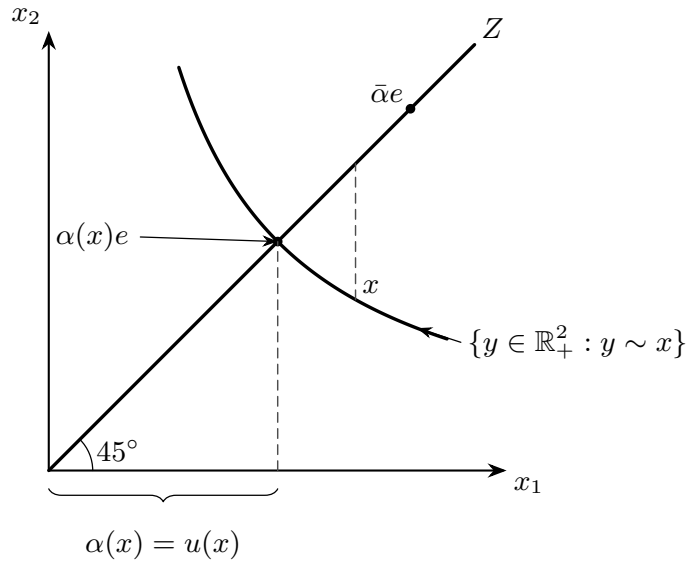
Continuity and utility function

A simple proof for monotone preference:

let e a vector with all elements 1.

monotonicity implies $x \succsim 0$ and $\forall \hat{\alpha}$ that $\hat{\alpha}e >> x$, we have $\hat{\alpha}e \succsim x$.

by monotonicity and continuity, $\exists \alpha(x) \in [0, \hat{\alpha}]$ such that $\alpha(x)e \sim x$.



$\alpha(x)$ as a utility function and we have $\alpha(x) \geq \alpha(y) \Leftrightarrow x \succsim y$:

If $\alpha(x) \geq \alpha(y) \xrightarrow{\text{monotonicity}} \alpha(x)e \succsim \alpha(y)e \Rightarrow x \succsim y.$

If $x \succsim y \Rightarrow \alpha(x)e \sim x \succsim y \sim \alpha(y)e \xrightarrow{\text{monotonicity}} \alpha(x) \geq \alpha(y).$

For proof of continuity of $u(x)$ see the book.

Utility function

As we saw before $u(x)$ is not unique. For any strictly increasing f, $f(u(x))$ is also a utility representation.

The previous proposition shows there always exist some continuous utility functions, but not all of utility functions are continuous.

We normally like utility functions that are (twice) differentiable.

But not all utility functions are differentiable like Leontief preferences $x \succsim y \Leftrightarrow \min\{x_1, x_2\} \geq \min\{y_1, y_2\}$.

Monotonicity of preference result in increasing utility function.

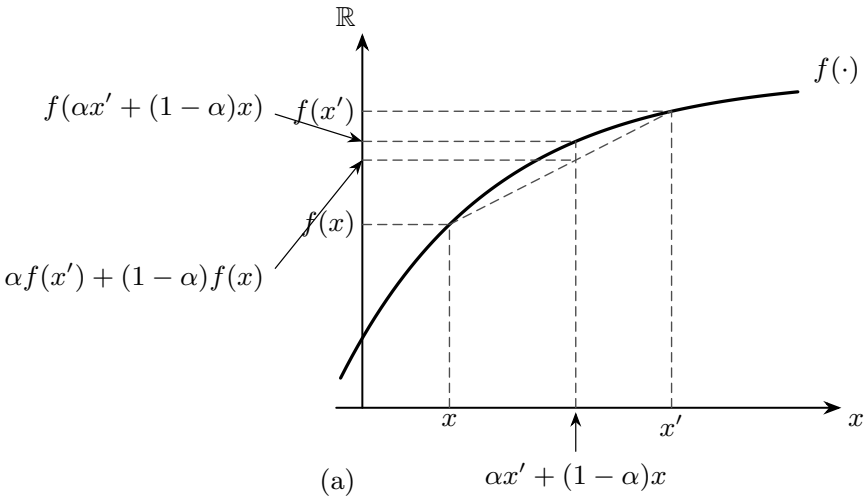
How convexity of preference is reflected in its utility functions?

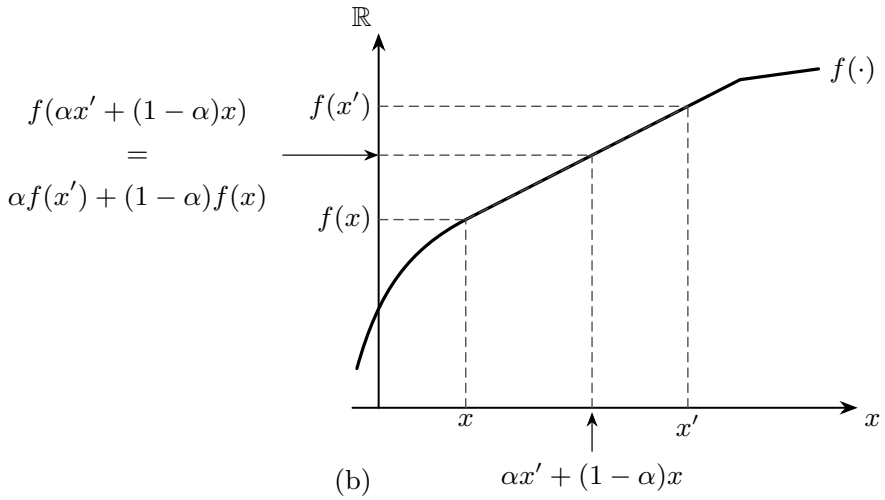
Concave functions

Definition (concavity)

$f : A \to \mathbb{R}$, where A is a convex set, is concave if $\forall x, x' \in A$ and $\forall \alpha \in (0,1)$: $f(\alpha x + (1-\alpha)x') \geq \alpha f(x) + (1-\alpha) f(x')$

If the inequality is strict when $x \neq x'$, f is strictly concave.

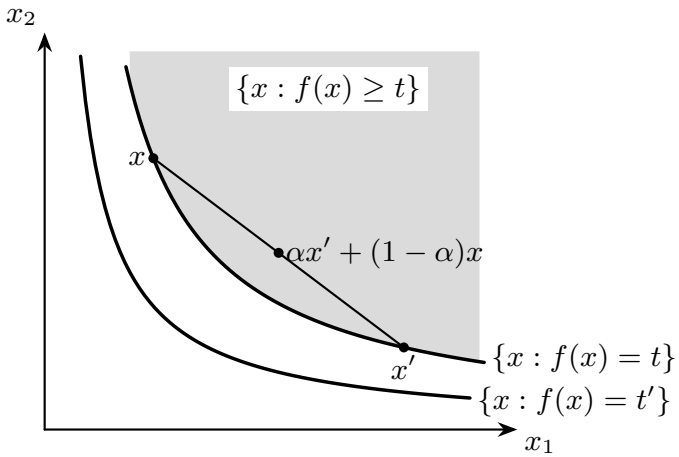


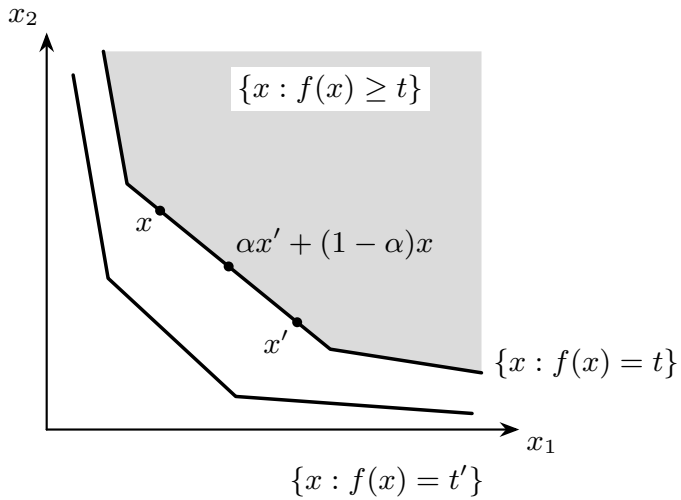


Quasi-concave functions

Definition (quasi-concavity)

$f : A \to \mathbb{R}$, where A is a convex set, is quasi-concave if its upper contour sets $\{x \in A : f(x) \geq t\}$ are convex.
If the contour sets are strictly convex, f is strictly quasi-concave.





Equivalent definition:
$\forall x, x' \in A$ and $\forall \alpha \in (0,1)$: $f(\alpha x + (1-\alpha)x') \geq \min[f(x), f(x')]$

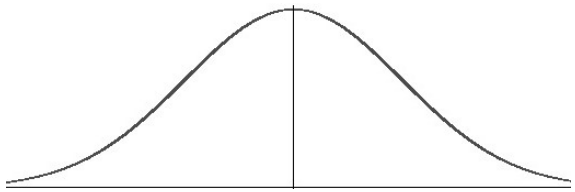
Concave vs. quasi-concave functions

Quasi-concavity is a weaker assumption than concavity.

A monotone function of one variable is quasi-concave (and quasi-convex).

Concavity is a cardinal property. It is not preserved under general transformation of an increasing function. Quasi-concavity is preserved.

Quasi-concavity relax unnecessary concavity in uncritical areas: where the function is monotone, it can be concave or convex, where it change slope, it must be concave.



Convex preference and utility function

Proposition

(a) If preference relation are represented by a concave function, it is convex. If it is represented by a strictly concave function, then it is strictly convex.
(b) If $u(.)$ is a utility function representing $\succsim$, then $u(.)$ is quasi-concave if and only if $\succsim$ is convex. $u(.)$ is strictly quasi-concave if $\succsim$ is strictly convex.

If $\succsim$ is convex, can we show that there is at least one concave utility function representing it?

Properties of utility functions

	if $\succsim$ is	$u(x)$ is
ordinal		
	monotone	increasing
	strictly monotone	strictly increasing
	convex	quasi-concave
	strictly convex	strictly quasi-concave
cardinal		
	homothetic	homogeneous of degree one
	quasilinear	linear in numeraire good

Ordinal properties are valid for all utility functions.

Cardinal properties are valid for at least one utility function.

Common utility functions

Linear (perfect substitute): $u(x) = \sum_i \alpha_i x_i$.

Leontief (perfect complement): $u(x) = \min\{\alpha_1 x_1, \ldots, \alpha_L x_L\}$

Cobb-Douglas: $u(x) = \sum_i x_i^{\alpha_i}$

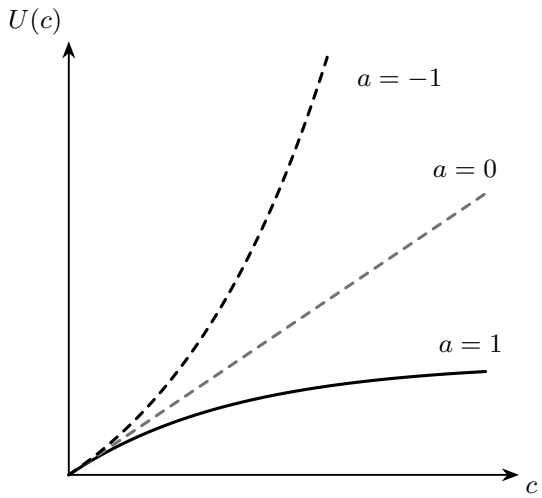
Constant elasticity of substitution (CES) is the general form of homothetic functions: $u(x) = \left(\sum_i \alpha_i x_i^{1/r} \right)^r$.

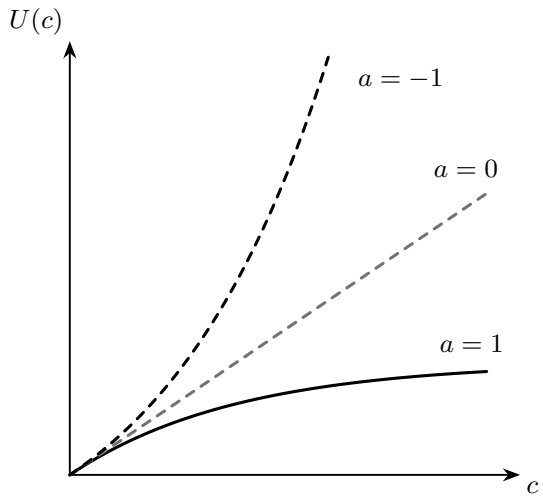
- If $r = 1$ we have a linear function;
- If r approaches zero, we get the Cobb–Douglas function;
- If r approaches negative infinity, we get the Leontief function.

Common utility functions

Quasi-linear utility: $u(x) = \alpha_1 x_1 + \theta(x_2, \ldots, x_L)$.

CARA utility: $u(x) = \dfrac{1 - e^{-ax}}{a}$, CRRA utility: $u(x) = \dfrac{x^{1-\eta} - 1}{1 - \eta}$





Other functions: HARA, Gorman polar, Uzawa, Stone-Geary, Greenwood-Hercowitz-Huffman, King-Plosser-Rebelo, etc.

The utility maximization problem (UMP)

We assume the consumer has rational, continuous, and locally nonsatiated preference

$$\max_{x \geq 0} u(x) \quad \text{s.t. } p.x \leq w$$

Proposition

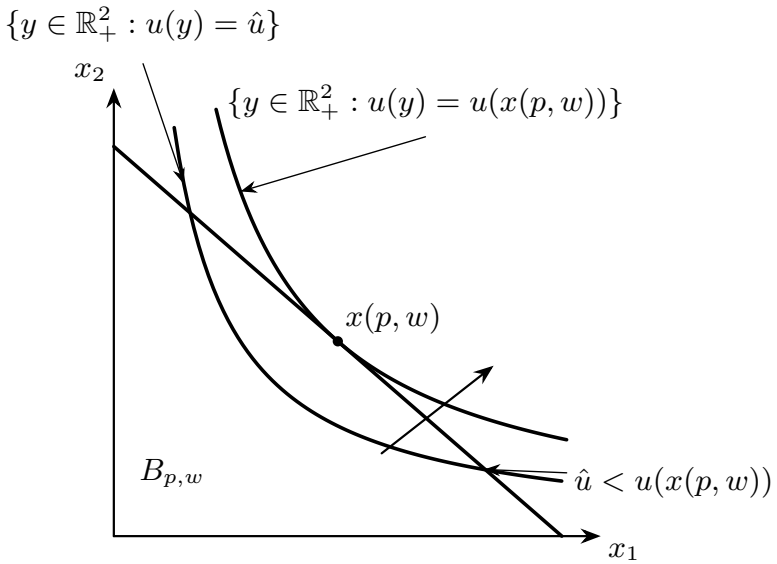
If $p >> 0$ and $u(.)$ is a continuous, then the utility maximization problem has a solution.

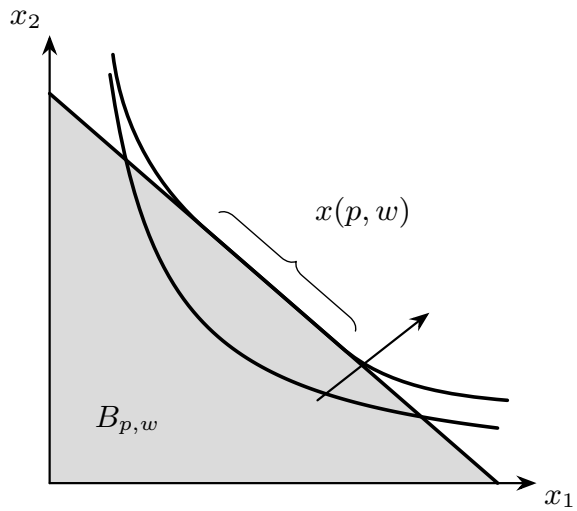
Proof: when $p >> 0$, the budget set is compact (bounded and closed) and a continuous function on a compact set always has a minimum and a maximum.

Walrasian (Marshallian) demand function

In chapter 2, we defined Walrasian demand in a general manner as $x(p, w)$ and imposed homogeneity of zero and Walras' law.

Here, we define Walrasian (Marshallian) demand as the set of optimal choices for each price-wealth as $x(p, w)$.





Properties of Walrasian demand

Proposition

Suppose that $u(.)$ is a continuous utility function of a locally nonsatiated preference on $\mathbb{R}^+$. Then the Walrasian demand correspondence $x(p, w)$ possesses the following properties:

(i) *homogeneity of degree zero in (p, w),*

(ii) *Walras's law $(p.x = w)$*

(iii) *if $\succsim$ is (strictly) convex, i.e. $u(.)$ is (strictly) quasi-concave, $x(p, w)$ is a (strictly) convex set.*

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(i) $\{x \in \mathbb{R}^+ : \alpha p.x \leq \alpha w\} = \{x \in \mathbb{R}^+ : p.x \leq w\}$

(ii) from local nonsatiation, if $p.x < w$, there is a y near x that $p.y < w$ and $y \succ x$, which contradicts x being optimal.

(iii) $x_1, x_2 \in x(p,w), \alpha \in [0,1], x_3 = \alpha x_1 + (1-\alpha)x_2$.
from quasi-concavity: $u(x_3) \geq u(x_1) = u(x_2) \rightarrow x_3$ is optimal.
also, $p.x_3 \leq w \rightarrow x_3$ is feasible. $\Rightarrow x_3 \in x(p,w)$.

Kuhn-Tucker conditions

If $u(x)$ is differentiable finding the optimum is much easier.

Kuhn-Tucker conditions say that if x^* is a solution of UMP, then there exist a Lagrange multiplier $\lambda \geq 0$ such that

$$\frac{\partial u(x^*)}{\partial x_l} \leq \lambda p_l, \text{ with equality if } x_l^* > 0$$

in gradient form

$$\nabla u(x^*) \leq \lambda p \text{ , and } \ x^*.[\nabla u(x^*) - \lambda p] = 0$$

if we are at an interior optimum

$$\nabla u(x^*) = \lambda p$$

A manual for constraint optimization

We have constrained maximization (minimization) with inequality constraints.

$$\max_x \ (\min_x) \ f(x) \text{ s.t. } g_1(x) \leq c_1, \ g_2(x) \leq c_2, \ldots$$

- Sort all inequality constraint to become $C_i(x) \geq 0$ ($C_i(x) \leq 0$).
- Write Lagrangian as $L(x, \lambda) = f(x) + \sum_i \lambda_i C_i$.
- Derive the FOC conditions
- Apply the complementarity slackness conditions

$$C_i \geq 0, \ \lambda_i \geq 0, \ \lambda_i C_i = 0$$

Deriving Kuhn-Tucker conditions

In constrained maximization (minimization), when we have an inequality constraint, write it as $C \geq 0$ ($C \leq 0$) and then the Lagrangian as $U(x) + \lambda C$.

The Lagrangian of UMP is written as $U(x) + \lambda(w - px)$.

Deriving Kuhn-Tucker conditions

In constrained maximization (minimization), when we have an inequality constraint, write it as $C \geq 0$ ($C \leq 0$) and then the Lagrangian as $U(x) + \lambda C$.

The Lagrangian of UMP is written as $U(x) + \lambda(w - px)$.

Note that we are also constrained by $x_l > 0$, so the true Lagrangian of the UMP is $U(x) + \lambda(w - px) + \sum_i \mu_i x_i$

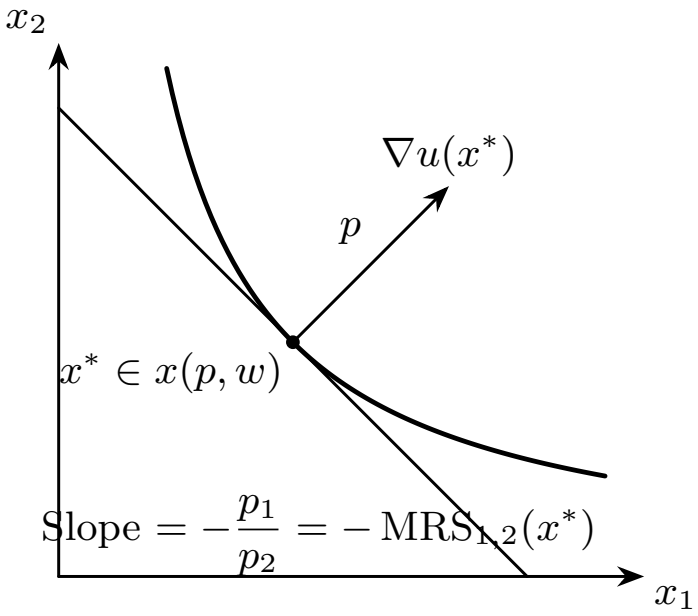
If a constraint is binding its associated Lagrangian multiplier is zero. Thus we have,

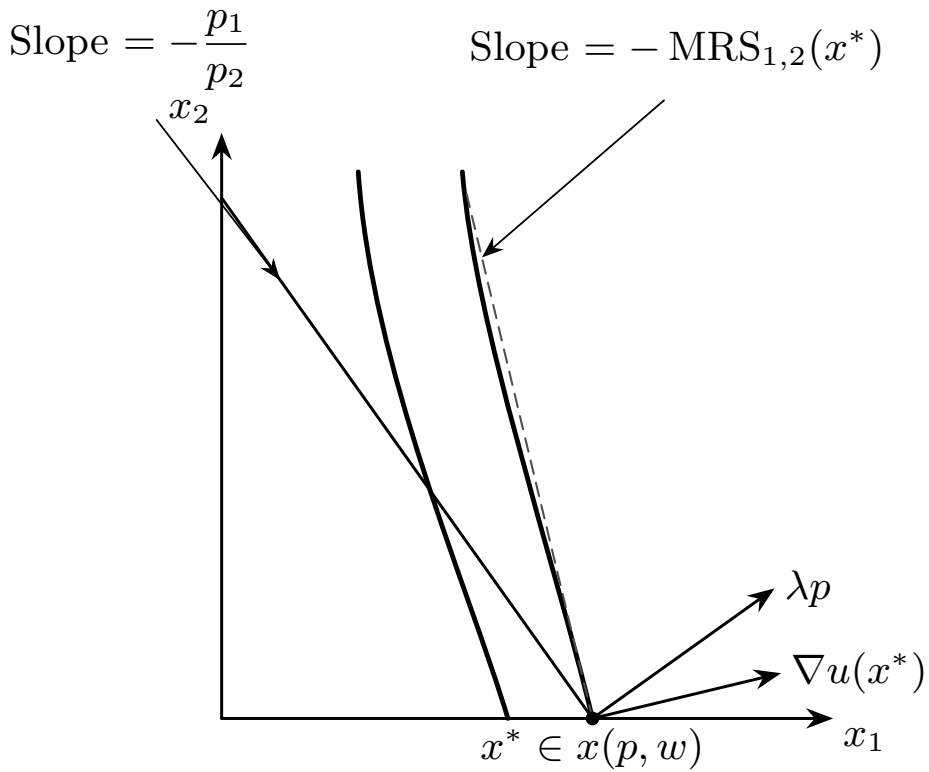
$$\lambda(w - p.x) = 0 \text{ and } \forall i,\ \mu_i x_i = 0$$

FOC w.r.t x_i: $\dfrac{\partial u(x^*)}{\partial x_i} - \lambda p_i + \mu_i = 0 \Rightarrow \dfrac{\partial u(x^*)}{\partial x_i} + \mu_i = \lambda p_i$

- if $x_i > 0 \rightarrow \mu_i = 0,\ \partial u(x^*)/\partial x_i = \lambda p_i,$
- if $x_i = 0 \rightarrow \mu_i > 0,\ \partial u(x^*)/\partial x_i < \lambda p_i$

Binding vs. unbinding constraint



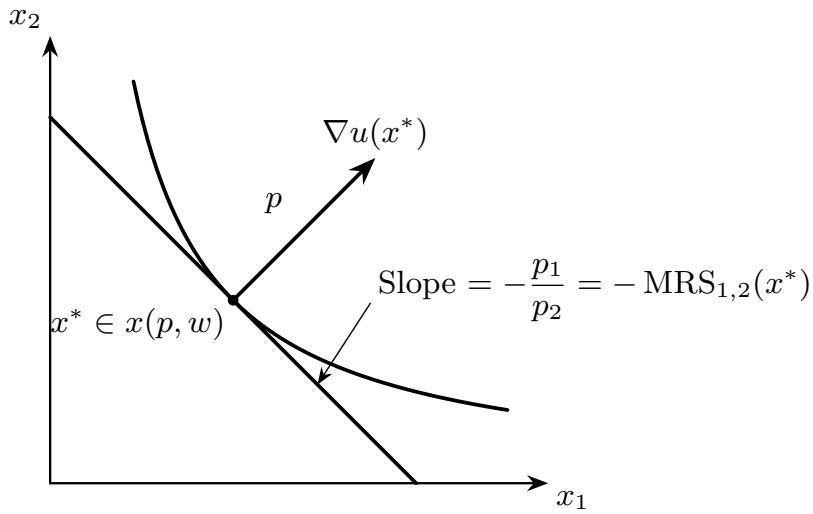


Marginal rate of substitution (MRS)

At an interior optimum, we have

$$\frac{\partial u/\partial x_i}{\partial u/\partial x_j} = \frac{p_i}{p_j}$$

At an interior optimum, MRS of any two goods is equal to their price ratio.



If $\frac{\partial u/\partial x_i}{\partial u/\partial x_j} > \frac{p_i}{p_j}$, increasing i with dx_i and decreasing j with $dx_i p_i/p_j$ is feasible and increases utility.

If we are at a corner and not an interior optimum this condition is not necessarily valid.

Lagrange mutliplier

The Lagrange multiplier, in general, shows the marginal (shadow) value of relaxing the constraint.

In UMP, the constraint is wealth, so λ shows the marginal utility of wealth:
Change in utility from a marginal increase in w :
$\nabla u(x(p,w)).D_w x(p,w)$,

we also have $\nabla u(x(p,w)) = \lambda p$

from Walras' law $p.x(p,w) = w \rightarrow p.D_w x(p,w) = 1$

Therefore,

$$\nabla u(x(p,w)).D_w x(p,w) = \lambda p.D_w x(p,w) = \lambda$$

Continuity and differentiability of Walrasian demand

When the Walrasian demand function is continuous?
If $\succsim$ is contiuous, strictly convex, and locally non-satiated.

When the Walrasian demand function is differentiable?
If and only if the determinant of *bordered Hessian* of $u(x)$ is non-zero.

$$\begin{vmatrix} D^2u(x) & \nabla u(x) \\ (\nabla u(x))^T & 0 \end{vmatrix} \neq 0$$

If $x(p, w)$ is differentiable, the Slutsky matrix $S(p, w)$ has maximal possible rank $L - 1$.

Indirect utility function

$v(p, w) = u(x(p, w))$ is called the indirect utility function.

It provides a short-cut from price-wealth to utility has several application in the rest of the course (e.g. computing demand from Roy's identity, welfare evaluation).

Proposition

Suppose that $u(.)$ is a continuous and representing a locally nonsatiated preference on $\mathbb{R}^+$. Then $v(p, w)$ is

(i) *homogeneous of degree zero,*

(ii) *strictly increasing in w,*

(iii) *non-increasing in p_l,*

(iv) *quasiconvex: $\forall v : \{(p, w) : v(p, w) \leq \bar{v}\}$ is convex,*

(v) *continuous in p and w*

The proof of (i) and (v) are similar to previous propositions.

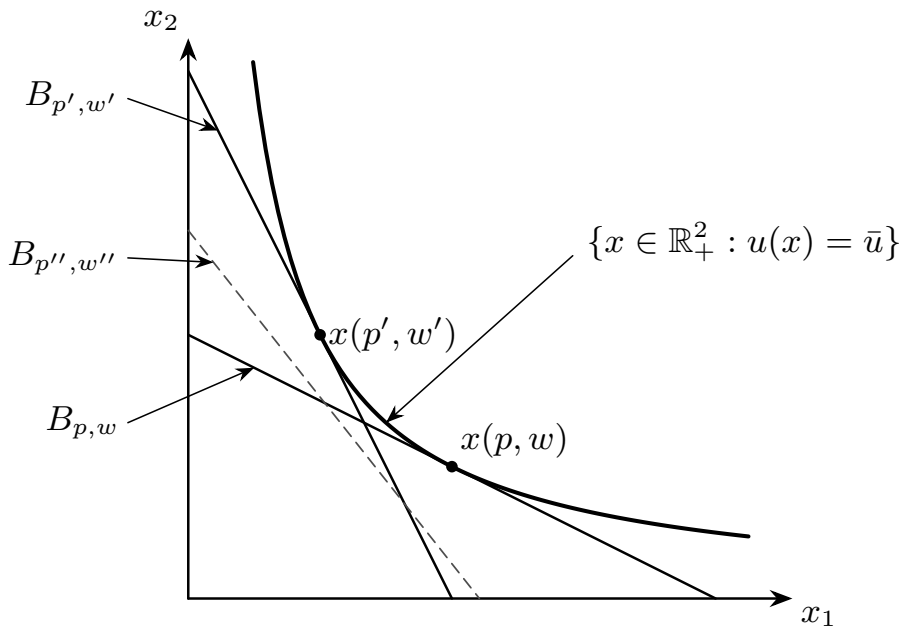
Indirect utility function

(ii) $\partial v/\partial w = \nabla u(x(p,w)).D_w x(p,w) = \lambda p.D_w x(p,w) = \lambda > 0$

(iii) Let $p_1 \leq p_0$ and $x_0 = x(p_0, w)$ then $(p_0 - p_1)x_0 \geq 0 \rightarrow$ $p_1 x_0 \leq p_0 x_0 \leq w$. Thus, x_0 is feasible under p_1 and $v(p_1, w) \geq u(x_0) = v(p_0, w)$.

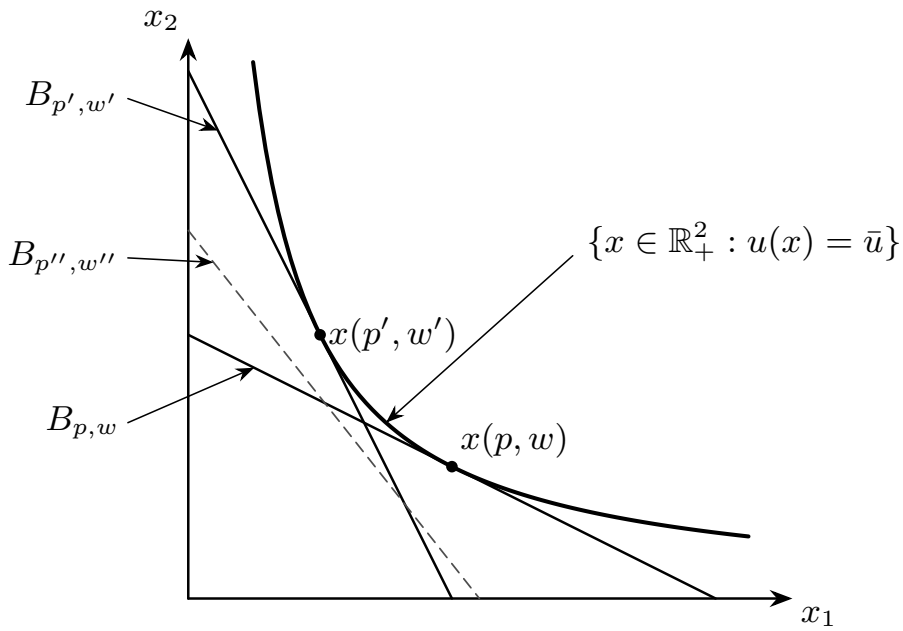
(iv) Suppose $v(p,w) \leq \bar{v}$, $v(p',w') \leq \bar{v}$, and for $\alpha \in [0,1]$ define $(p'', w'') = (\alpha p + (1-\alpha)p', \alpha w + (1-\alpha)w')$.
For x that $p''.x \leq w''$ either $p.x \leq w$ or $p'.x \leq w'$ or both. Whichever of the inequalities holds, we can conclude that $v(p'', w'') \leq \bar{v}$.

Indirect utility function



$v(p, w)$ is not graph-friendly, why?

Indirect utility function



$v(p, w)$ is not graph-friendly, why? at least 3 arguments p_1, p_2, w.
What does quasiconvexity mean for the contour sets?

If $\hat{u}(x) = f(u(x))$, then $\hat{v}(p, w) = f(v(p, w))$.

Expenditure minimization problem (EMP)

EMP is the dual problem of the UMP.

$$\min_{x \geq 0} \; p.x \qquad \text{s.t. } u(x) \geq u$$

Proposition

Suppose $u(.)$ is a continuous utility function of a locally nonsatiated preference on $\mathbb{R}^+$ and the price vector is $p >> 0$. We have

(i) *if x^* is optimal in the UMP when wealth is $w > 0$, then x^* is optimal in the EMP when the required utility level is $u(x^*)$. Moreover, the minimized expenditure level in the EMP is exactly w.*

(ii) *If x^* is optimal in the EMP when the required utility level is $u > u(0)$, then x^* is optimal in the UMP when wealth is $p.x^*$. Moreover, the maximized utility level in this UMP is exactly u.*

EMP

Proof.

(i) if x^* is not optimal in the EMP, then $\exists x', u(x') \geq u(x^*)$ and $p.x' < p.x^* \leq w$. Then, local nonsatiation: $\exists x'', u(x'') > u(x')$ and $p.x'' < w \Rightarrow u(x'') > u(x^*) \Rightarrow x^*$ is not optimal in the UMP. By Walras law: $p.x^* = w$.

(ii) if x^* is not optimal in UMP then $\exists x', u(x') > u(x^*)$ and $p.x' \leq p.x^*$. $\exists \alpha \in (0,1), x'' = \alpha x', u(x'') > u(x^*)$ and $p.x'' < p.x^* \Rightarrow x^*$ is not optimal in EMP.

The expenditure function

$e(p, u) = p.x^*$ is the expenditure function, where x^* is solves the EMP.

It has several applications like recovering preference from demand (integrability), welfare evaluation, etc.

Proposition

Suppose $u(.)$ is a continuous representing locally nonsatiated preference on $\mathbb{R}^+$. Then, $e(p, u)$ is

i. *homogeneous of degree one in p.*
ii. *strictly increasing in u and $\forall l$ nondecreasing in p_l.*
iii. *concave in p.*
iv. *continuous in p and u.*

The expenditure function

Proof:

(i) price change doesn't change the constraint in the EMP.
$\forall \alpha > 0, \min \alpha p.x$ gives a same x^* as $\min p.x$. Thus,
$e(\alpha p, u) = \alpha p.x* = \alpha e(p, u)$.

(ii) If $e(p, u)$ is not strictly increasing in u, we have $u'' > u'$ when $p.x' \geq p.x'' > 0$. By continuity of $u(.)$, $\exists \alpha \in (0, 1), \tilde{x} = \alpha x''$ that $u(\tilde{x}) > u'$ and $p.x' > p.\tilde{x} \Rightarrow x'$ is not optimal in the EMP for u'.

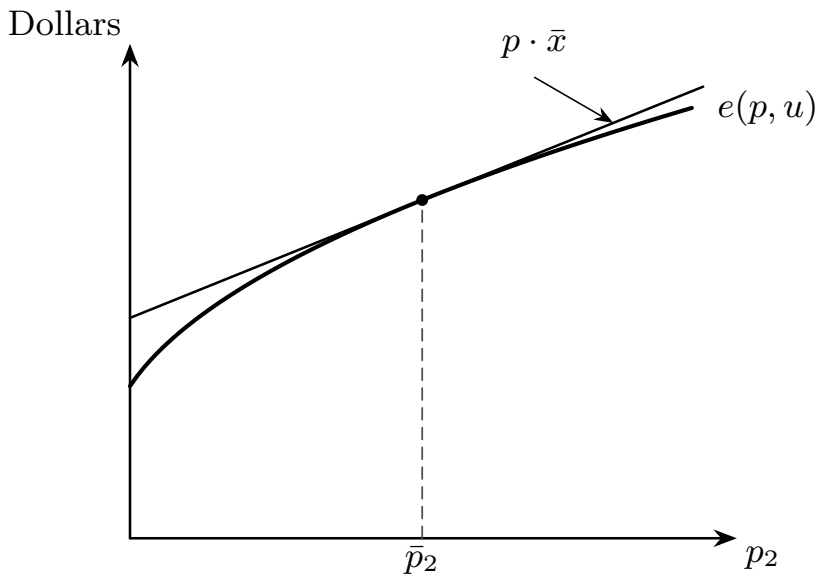
Assume p' equals p'' except in $p''_l \geq p'_l$. Then, we have
$e(p'', u) = p''.x'' \geq p'.x'' \geq e(p', u)$.

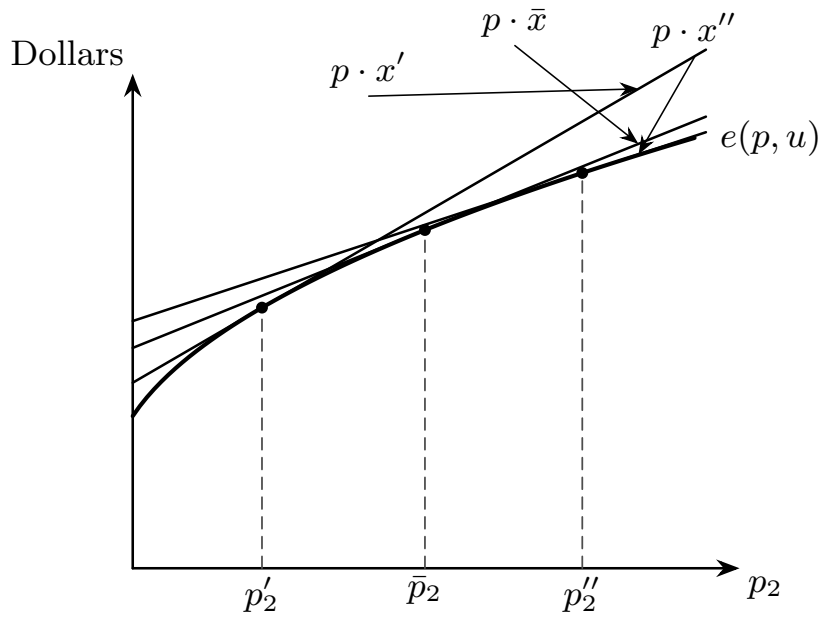
(iii) Assume $p'' = \alpha p + (1 - \alpha)p'$. Then, for a fix utility $\bar{u}$

$$e(p'', \bar{u}) = p''.x'' = \alpha p.x'' + (1 - \alpha)p'.x'' \geq \alpha e(p, \bar{u}) + (1 - \alpha)e(p', \bar{u})$$

Why the expenditure function is concave in p?

$e(p, u) = p.x$ is linear in p. If price changes, with a fixed bundle $\bar{x}$, we move along the line $p.\bar{x}$. The minimized expenditure must be below $p.\bar{x}$ for other values of p.





Expenditure function vs. indirect utility function

for any $p >> 0, w > 0$, and $u > u(0)$ we have

$$e(p, v(p, w)) = w, \qquad v(p, e(p, u)) = u$$

For a fixed price, $e(p, .)$ and $v(p, .)$ are the inverse of each other.

Hicksian demand

The set of optimal commodity vectors in the EMP build $h(p,u) \subset \mathbb{R}^+$ and is called Hicksian or compensated demand.

Proposition

Suppose that $u(.)$ is a continuous utility function of a locally nonsatiated preference on $\mathbb{R}^+$. The Hicksian demand has these properties:

(i) *homogeneity of degree zero in price: $h(\alpha p, u) = h(p,u)$*

(ii) *no excess utility: $\forall x \in h(p,u), u(x) = u$.*

(iii) *convexity/uniqueness: If $\succsim$ is convex, then $h(p,u)$ is a convex set. If $\succsim$ is strictly convex, $h(p,u)$ has a unique element.*

(i) $\min p.x$ and $\min \alpha p.x$ gives same x for $\alpha > 0$.
(ii) If $x \in h(p,u)$ and $u(x) > u$ then by continuity of $u(.)$,
$\exists \alpha \in (0,1), x' = \alpha x : u(x') \geq u$ and $p.x' < p.x \Rightarrow x$ is not optimal.

Kuhn-Tucker conditions for Hicksian demand

$$\min_{x \geq 0} \; p.x, \qquad \text{s.t. } u(x) \geq u$$

Kuhn-Tucker conditions for Hicksian demand

$$\min_{x \geq 0} \; p.x, \qquad \text{s.t. } u(x) \geq u$$

Lagrangian: $p.x + \lambda(u - u(x)) + \sum_i \mu_i x_i$

F.O.C. w.r.t. x_i : $\quad p_i - \lambda \partial u / \partial x_i + \mu_i = 0$

if $x_i > 0 \rightarrow \mu_i = 0,\ p_i = \lambda \partial u / \partial x_i$.
if $x_i = 0 \rightarrow \mu_i > 0$ and $p_i > \lambda \partial u / \partial x_i$

Therefore, $\lambda \nabla u(x^*) \leq p$ and $x^*.[p - \lambda \nabla(x^*)] = 0$ for $\lambda \geq 0$.

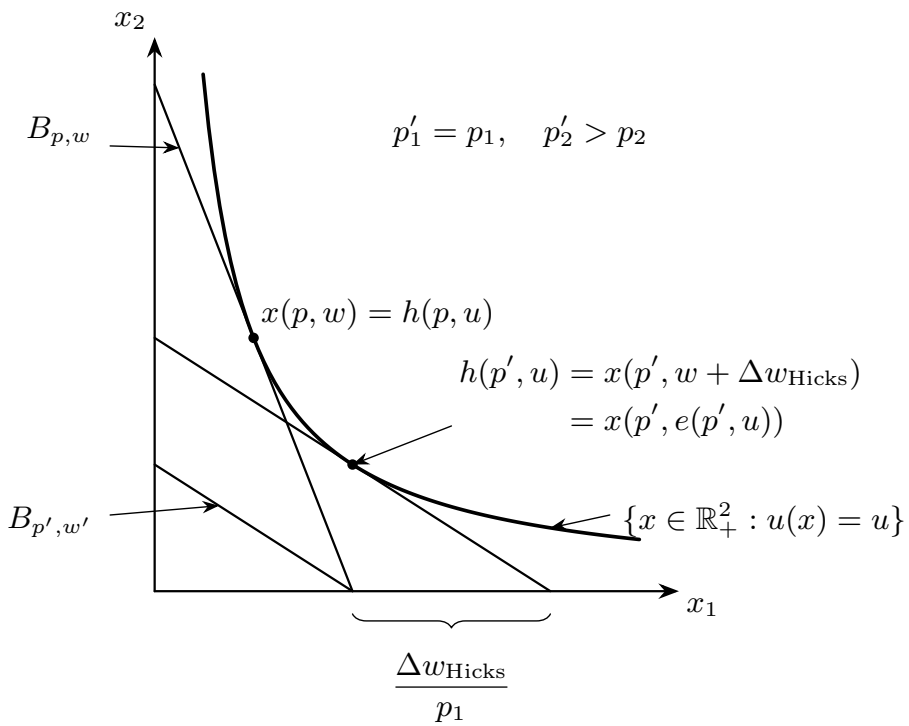
What is the marginal cost of utility?

Hicksian vs. Walrasian demand

$$h(p,u) = x(p, e(p,u)), \qquad x(p,w) = h(p, v(p,w))$$

With a change in prices, $h(p,u)$ gives the demand if the consumer is "compensated" for her wealth to keep her initial utility level.

Hicksian wealth compensation: $\Delta w_{\text{Hicks}} = e(p',u) - w$



Compensated law of demand

Demand and price move in opposite directions.

Proposition

For continuous $u(.)$, locally nonsatiated $\succsim$, with single-valued $h(p,u)$

$$(p'' - p').[h(p'', u) - h(p', u)] \leq 0$$

We have

$$p''.h(p'', u) \leq p''.h(p', u)$$

and

$$p'.h(p'', u) \geq p'.h(p', u)$$

Subtracting these give the result.

Thus, no Giffen good under Hicksian demand.

Example: Cobb-Douglas utility

$$\max_{x_1,x_2} \; x_1^{\alpha} + x_2^{(1-\alpha)} \qquad \text{s.t. } p_1x_1 + p_2x_2 = w$$

$$x_1(p,w) = \frac{\alpha}{p_1}w, \qquad x_2(p,w) = \frac{1-\alpha}{p_2}w$$

$$v(p,w) = \ln \alpha^{\alpha}(1-\alpha)^{1-\alpha} + \ln w - \alpha \ln p_1 - (1-\alpha)\ln p_2$$

$$h_1(p,u) = \Big(\frac{\alpha p_2}{(1-\alpha)p_1}\Big)^{1-\alpha}u, \qquad h_2(p,u) = \Big(\frac{(1-\alpha)p_1}{\alpha p_2}\Big)^{\alpha}u$$

$$e(p,u) = [\alpha^{-\alpha}(1-\alpha)^{\alpha-1}]p_1^{\alpha}p_2^{1-\alpha}u$$

Duality theory

Duality theorem is a result of two properties in microeconomics: convexity and linearity of prices.

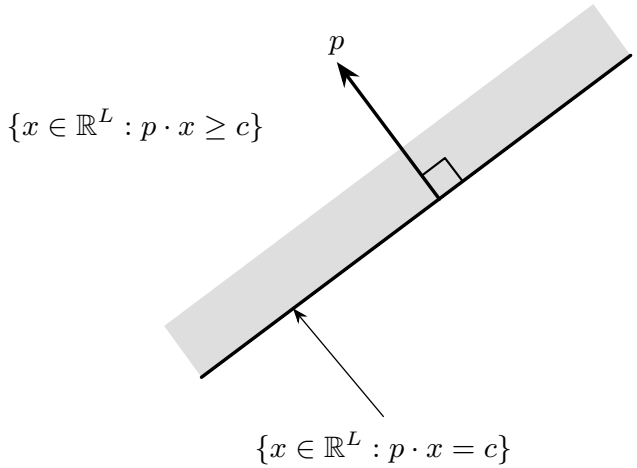
We have duality in consumer and producer theories.

- consumer: UMP and EMP
- producer: cost and profit functions (Hotelling's and Shephard's lemmas)

Essentially, closed convex sets can be described in two ways:

- listing their elements
- listing half-spaces that contains it.

what is half-space?

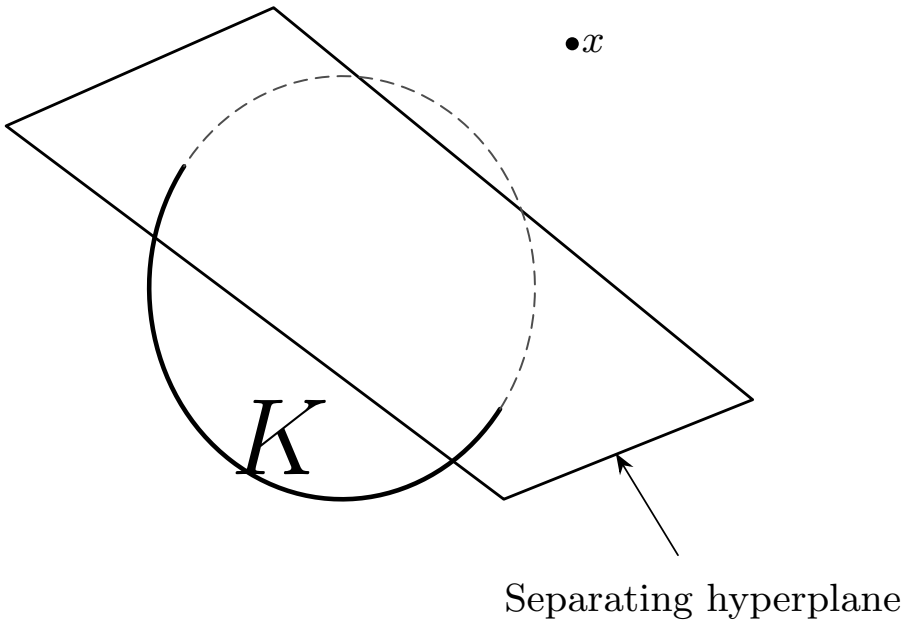


Half-spaces

A half-space is a set of the form $\{x \in \mathbb{R}^L : \ p.x \geq c\}$ for some nonzero $p \in \mathbb{R}^L$ and $c \in \mathbb{R}$.

The boundary of the half-space is the hyperplane $\{x \in \mathbb{R}^L : \ p.x = c\}$ and p is the vector prepedicular to the border hyperplane of the half-space.

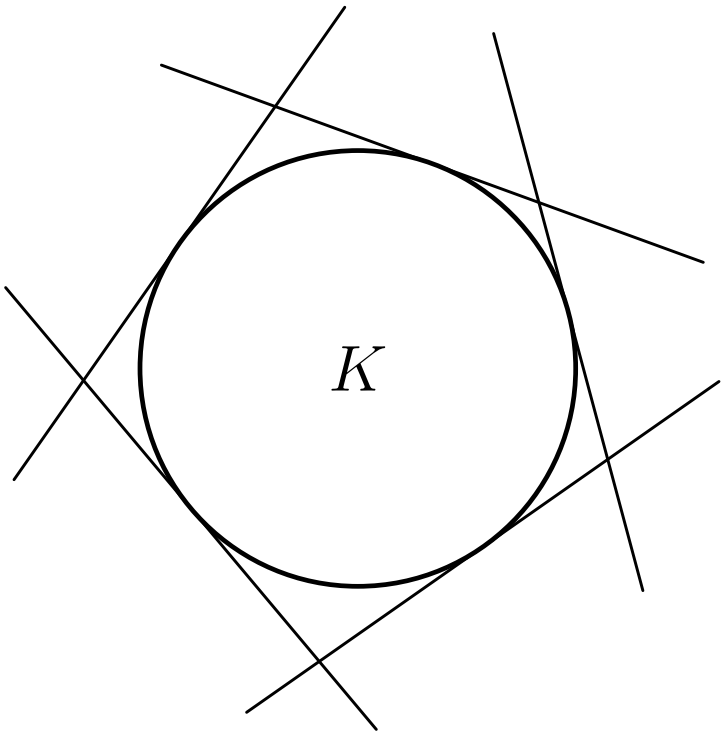
Separating hyperplane theorem: if K is a convex and closed set (contains its border), and x is not on K, then there is a half space including K and excluding x.

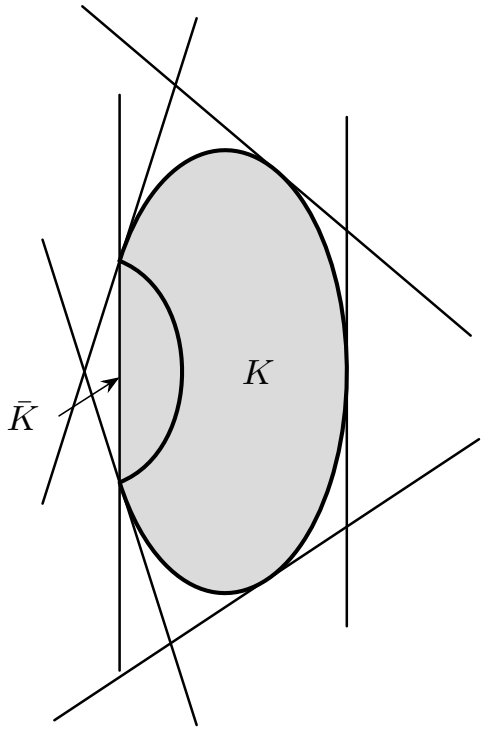


Convex set from half-spaces

We can build all convex and closed sets by the intersection of the half-spaces containing it.

Even if the set is not convex, the intersection of half-spaces gives the smallest closed, convex set containing K which is called *convex hull* of K.





Support function

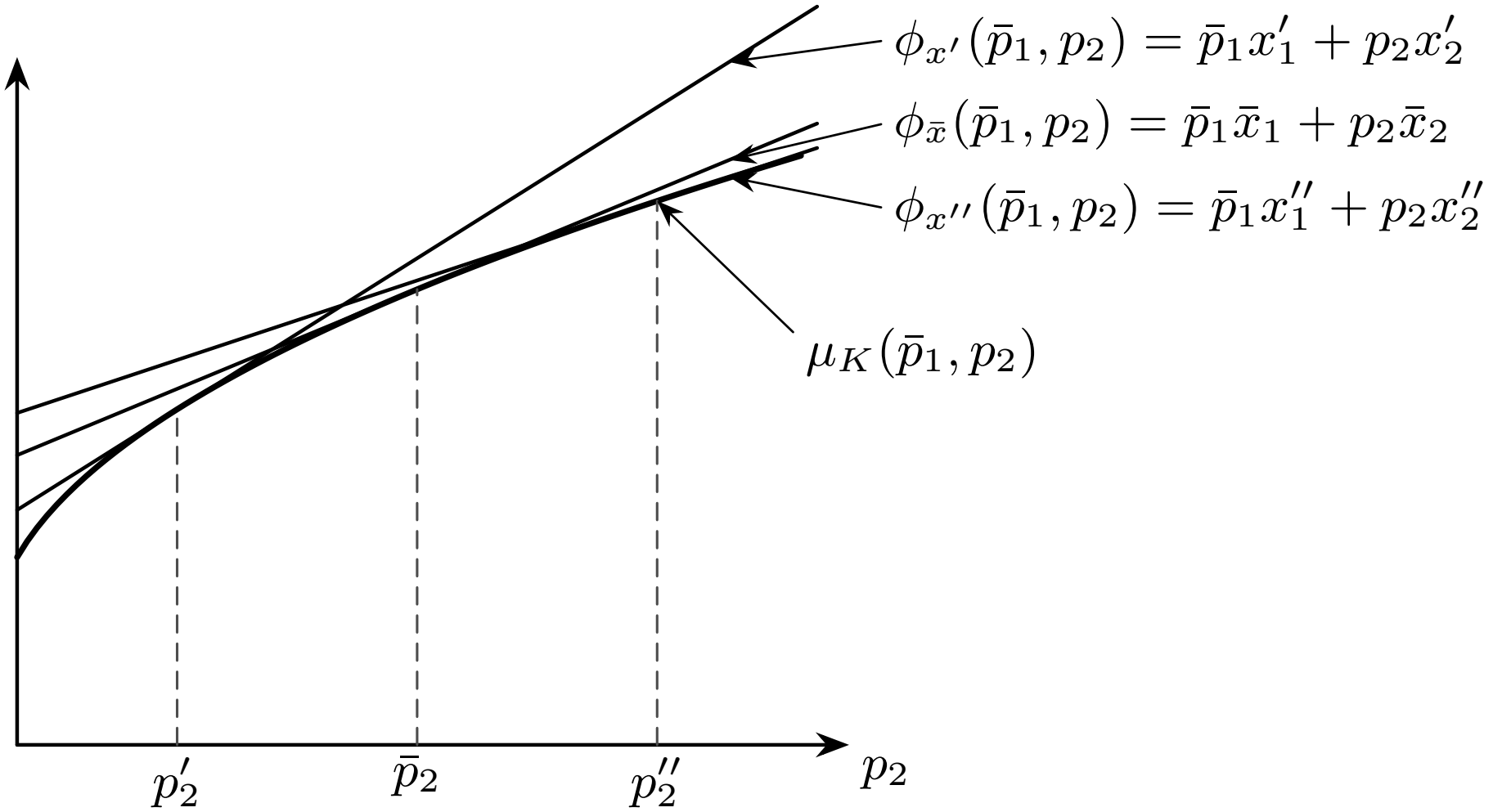
The support function of a closed non-empty set K is defined for any $p \in \mathbb{R}^L$ as $\mu_K(p) = \inf\{p.x : x \in K\}$

infimum is the minimum but may not be in the set.
When K is convex, $\mu_K(.)$ provides a "dual" description of the set K, because we can rebuild K from knowledge of $\mu_K(.)$:

$$K = \{x \in \mathbb{R}^L : \ \forall p, \ p.x \geq \mu_K(p)\}$$

μ_K is homogeneous of degree one and concave (which function in the lecture is similar to this?)

Concavity of support function



The duality theorem

Proposition

Let K a nonempty closed set and $\mu_k(.)$ its support function. There is a unique $\bar{x} \in K$ such that $\bar{p}.\bar{x} = \mu_k(\bar{p})$ if and only if $\mu_k(.)$ is differentiable at $\bar{p}$. in this case $\nabla \mu_k(\bar{p}) = \bar{x}$.

The expenditure function is the support function of the set $\{x \in \mathbb{R}^L_+ : \ u(x) \geq x\}$.

Thus, the properties of a support function, such as homogeneity of degree zero and concavity would immediately follow.

Hicksian demand and expenditure function

$e(p,u) = p.h(p,u)$

Proposition

$u(.)$ is a continuous utility function of a locally nonsatiated and strictly convex preference on $\mathbb{R}^L_+$.

$$h(p,u) = \nabla_p e(p,u) \text{ or } \forall\ l: \ h_l(p,u) = \frac{\partial e(p,u)}{\partial p_l}$$

Proof 1. Duality theorem.

Proof 2. FOC: $\nabla_p e(p,u) = h(p,u) + [p.D_p h(p,u)]^T$. For an interior optimum, we have $p = \lambda \nabla_p u(h(p,u))$, thus:
$\nabla_p e(p,u) = h(p,u) + \lambda[\nabla_p u(h(p,u)).D_p h(p,u)]^T$, but from EMP we have $u(h(p,u)) = u$ and thus $\nabla_p u(h(p,u)).D_p h(p,u) = 0$.

Proof 3. Envelope theorem.

Envelope theorem

The Envelope theorem is a result about the differentiability properties of the objective function of a parameterized optimization problem.

$$\phi(\alpha) = \min_{x} \; f(x, \alpha), \quad \text{s.t. } g(x, \alpha) = 0, \text{ with the solution } x^*(\alpha)$$

$$\nabla_\alpha \phi(\alpha) = \nabla_\alpha f(x^*(\alpha), \alpha) + \lambda \nabla_\alpha g(x^*(\alpha), \alpha)$$

As we change parameters of the objective, changes in the optimizer of the objective do not contribute to the change in the objective function.

In the EMP price vector appears only in f and not g, thus $\nabla_p e(p, u) = h(p, u)$.

Derivative of Hicksian demand

Proposition

$u(.)$ is a continuous utility function of a locally nonsatiated and strictly convex preference on $\mathbb{R}^L_+$. Also, $h(p,u)$ is differentiable at (p,u). Then,

(i) $D_p h(p,u) = D_p^2 e(p,u)$

(ii) $D_p h(p,u)$ *is a negative semidefinite matrix.*

(iii) $D_p h(p,u)$ *is symmetric.*

(iv) $D_p h(p,u)p = 0$

Properties (i) to (iii) are the results of simple matrix calculus.

How to prove property (iv)?

Properties of Hicksian demand

Negative semi-definiteness of $D_p h(p,u)$ is equivalent to the compensated law of demand.

$$dp.D_p h(p,u)dp \leq 0 \rightarrow \ dp.dh(p,u) \leq 0$$

$D_p h(p,u)$ is symmetric, so compensated cross-price derivatives of two goods are equal:

$$\partial h_l(p,u)/\partial p_k = \partial h_k(p,u)/\partial p_l$$

What does this mean? why we have this property?

Properties of Hicksian demand

Negative semi-definiteness of $D_p h(p,u)$ is equivalent to the compensated law of demand.

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$$\partial h_l(p,u)/\partial p_k = \partial h_k(p,u)/\partial p_l$$

What does this mean? why we have this property? This is because transitivity in rational preferences makes it path-independent:

$x \succsim y$ we have the same result in another path $x \succsim z_1 \succsim z_2 \cdots \succsim y$

Substitutes and complements goods

Goods i and j at (p, u) are

- substitutes if $\partial h_l(p, u)/\partial p_k \geq 0$ and
- complements if $\partial h_l(p, u)/\partial p_k \leq 0$

For Walrasian demand we use term "gross" substitute/complements.

Gross substitute is not very useful since it does not have symmetry.

We had $D_p h(p, u)p = 0$ and $\partial h_l(p, u)/\partial p_l \leq 0$, so for at least one k, $\partial h_l(p, u)/\partial p_k \geq 0$ (each good has at least one substitute).

Hicksian vs. Walrasian demand functions

What is a demand function?

Hicksian vs. Walrasian demand functions

What is a demand function?
Quantity demanded for a good as a function of its price.

Prices are relative and helps to compare between different goods.

To determine demand, in addition to relative terms (prices), we need an overall term (utility or income)

We need to keep either income (observable) or utility (unobservable) constant.

Therefore, there are two types of demand functions:

- Walrasian (Marshalian): Total income is constant.
- Hicksian: The utility is constant

Hicksian vs. Walrasian demand functions

Utility is an abstract thing, why do we use it for defining an abstract demand function?

More importantly, Walrasian demand is observable demand function, but Hicksian is unobservable.

Some reasons to like Hicksian demand: It enables us to

- determine substitutability and complementarity of two goods.
- distinguish income from substitution effect of a price change
- in many theoretical problems, we have to use compensated demand of consumers (e.g. welfare evaluation, optimal taxation)

How can we estimate Hicksian demand?

Connecting Hicksian and Walrasian demand functions

Slutsky equation is our tool to find the unobservable Hicksian function to the observable Walrasian function.

Proposition

$u(.)$ is a continuous utility function of a locally nonsatiated and strictly convex preference on $\mathbb{R}^L_+$. Then for all (p, w) we have

$$\frac{\partial h_l(p,u)}{\partial p_k} = \frac{\partial x_l(p,w)}{\partial p_k} + \frac{\partial x_l(p,w)}{\partial w} x_k(p,w)$$

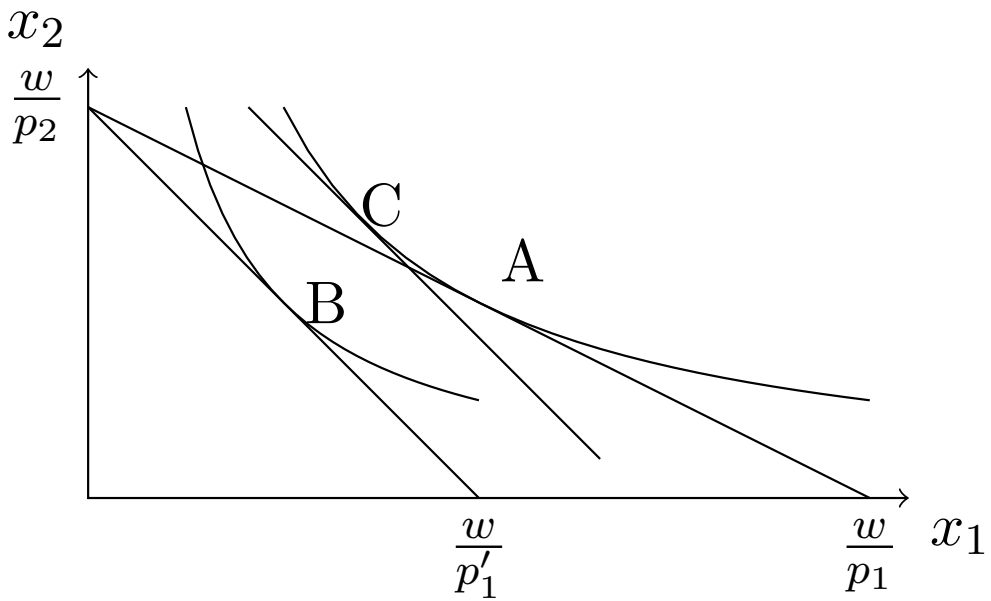
Proof. $h_l(p,u) = x_l(p,w)$ and $w = e(p,u)$

$$\frac{\partial h_l(p,u)}{\partial p_k} = \frac{\partial x_l(p,w)}{\partial p_k} + \frac{\partial x_l(p,w)}{\partial w} \frac{\partial e(p,u)}{\partial p_k}$$

We have $\frac{\partial e(p,u)}{\partial p_k} = h_k(p,u)$ and at $w = e(p,u)$, $h_k(p,u) = x_k(p,w)$.

Graphical interpretation of Slutsky equation

Change in demand after a price change Δp: $\underbrace{\Delta x_{AC}}_{\text{substitution effect (Hicksian demand)}} + \overbrace{\Delta x_{CB}}^{-\text{income effect}} = \underbrace{\Delta x_{AB}}_{\text{total effect (Walrasian demand)}}$



$$\Delta x_{AC} = \Delta x_{AB} + \Delta x_{BC} \Rightarrow \frac{\Delta x_{AC}}{\Delta p_1} = \frac{\Delta x_{AB}}{\Delta p_1} + \frac{\Delta x_{BC}}{\Delta w}\frac{\Delta w}{\Delta p_1}$$

$$p_1 x_1 + p_2 x_2 = w \Rightarrow \Delta w/\Delta p_1 = x_1, \text{ thus } \frac{\Delta x_{AC}}{\Delta p_1} = \frac{\Delta x_{AB}}{\Delta p_1} + \frac{\Delta x_{BC}}{\Delta w} x_1$$

Slutsky equation in terms of elasticities

$$\frac{\partial x_l(p,w)}{\partial p_k} = \underbrace{\frac{\partial h_l(p,u)}{\partial p_k}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_l(p,w)}{\partial w} x_k(p,w)}_{\text{income effect}}$$

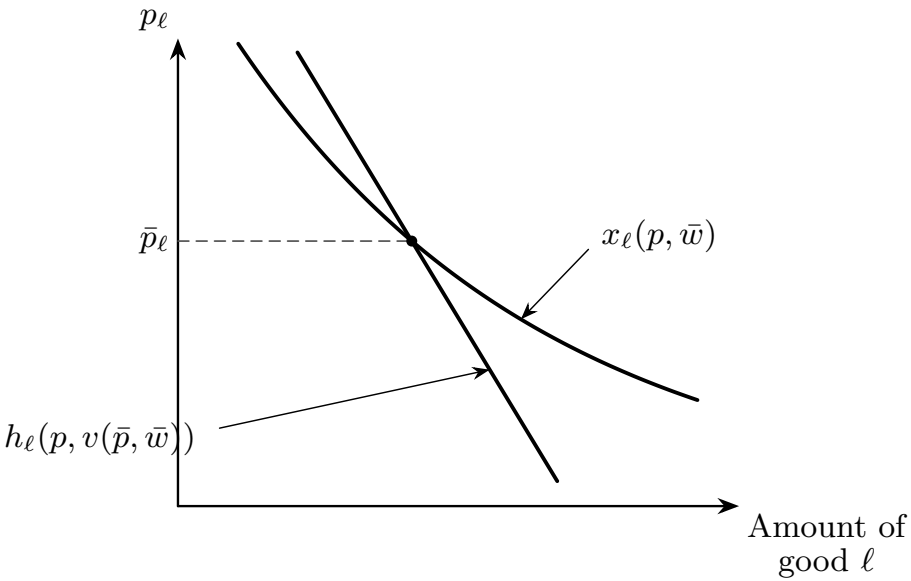
$$\xRightarrow{\times p_k/x_l} \quad \varepsilon^U_{lk} = \varepsilon^C_{lk} - \eta_l b_k$$

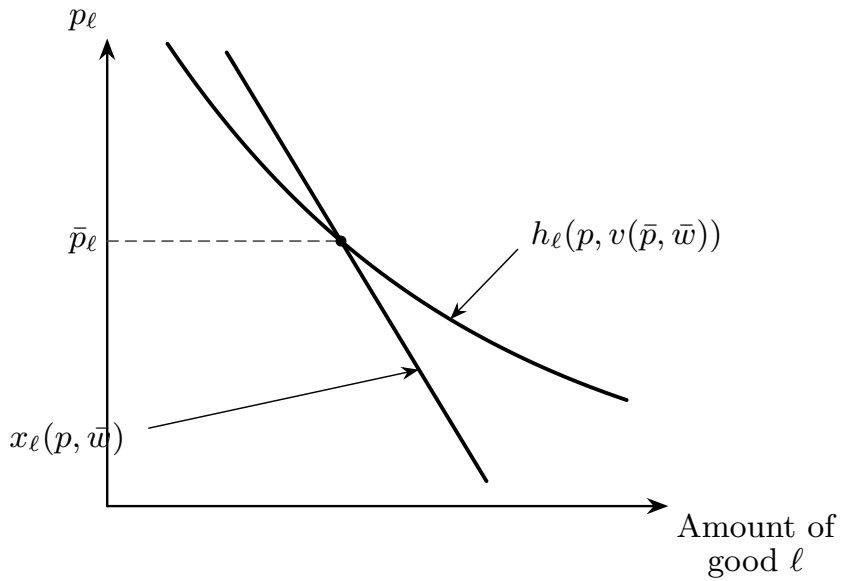
ε^U: elasticity of Walrasian (uncompensated) demand
ε^C: elasticity of Hicksian (compensated) demand
η: income elasticity of demand
b is budget share of the good.

For own-price Slutsky equation:

$$\varepsilon^U_{ll} = \varepsilon^C_{ll} - \eta_l b_l$$

Normal vs. inferior goods





Slutsky substitution matrix

The previous proposition tell us $D_p h(p,u) = S(p,w)$ where

$$S(p,w) = \begin{bmatrix} s_{11}(p,w) & \dots & s_{1L}(p,w) \\ \vdots & & \vdots \\ s_{L1}(p,w) & \dots & s_{LL}(p,w) \end{bmatrix}$$

with $s_{lk}(p,w) = \partial x_l(p,w)/\partial p_k + x_k(p,w)\partial x_l(p,w)/\partial w$.

What can we do with $S(p,w)$?

Slutsky substitution matrix

The previous proposition tell us $D_p h(p,u) = S(p,w)$ where

$$S(p,w) = \begin{bmatrix} s_{11}(p,w) & \dots & s_{1L}(p,w) \\ \vdots & & \vdots \\ s_{L1}(p,w) & \dots & s_{LL}(p,w) \end{bmatrix}$$

with $s_{lk}(p,w) = \partial x_l(p,w)/\partial p_k + x_k(p,w)\partial x_l(p,w)/\partial w$.

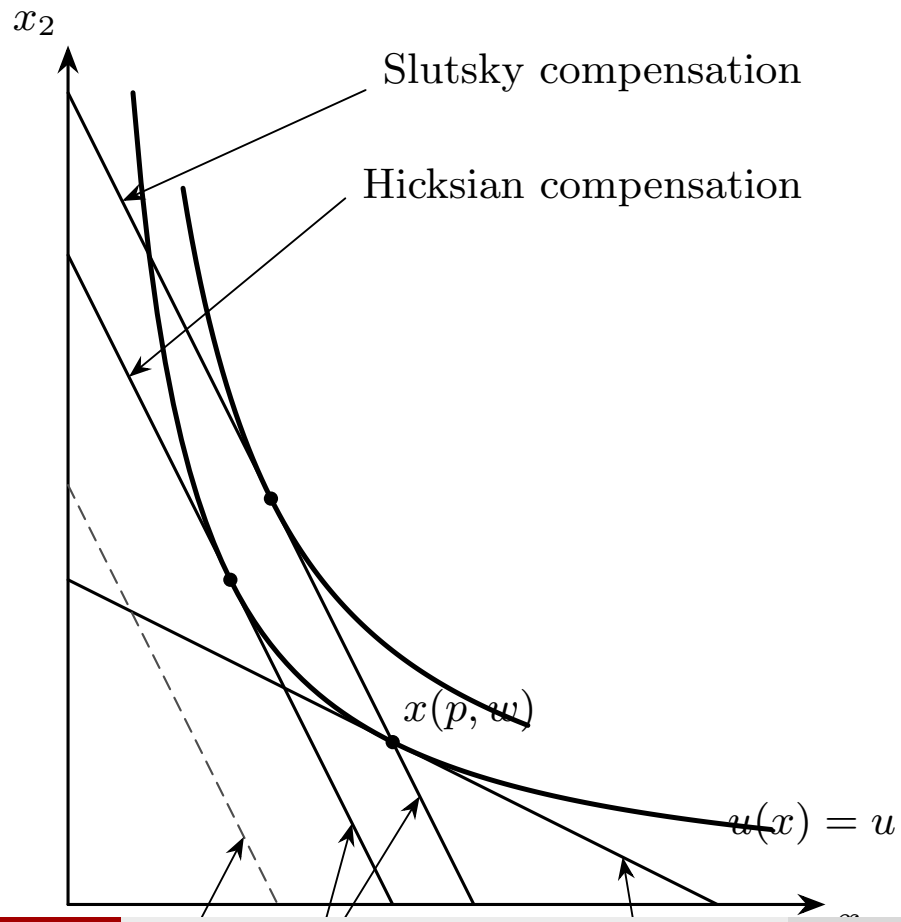
What can we do with $S(p,w)$?

If demand function is generated from a rational preference maximization, $S(p,w)$ must be negative semidefinite, symmetric and $S(p,w)P = 0$.

Hicksian vs. Slutskky compensation

$\Delta w_{\text{Slutsky}} = p'.x(\bar{p}, \bar{w}) - \bar{w}$ and $\Delta w_{\text{Hicks}} = e(p', \bar{u}) - \bar{w}$
$\Rightarrow \Delta w_{\text{Hicks}} \leq \Delta w_{\text{Slutsky}}$,

but in differentials: $dw_{\text{Slutsky}} = x(\bar{p}, \bar{w})$ and $dw_{\text{Slutsky}} = h(\bar{p}, \bar{w})$



Walrasian demand and indirect utility

From EMP, we had $h_l(p,u) = \partial e(p,u)/\partial p_l$, but why we don't have $x_l(p,u) = \partial v(p,u)/\partial p_l$?

Walrasian demand and indirect utility

From EMP, we had $h_l(p,u) = \partial e(p,u)/\partial p_l$, but why we don't have $x_l(p,u) = \partial v(p,u)/\partial p_l$? Because v is not unique and varies with a transformation of utility.

How to normalize v? Roy's identity

Proposition

$u(.)$ is a continuous utility representing a locally nonsatiated and strictly convex preference on $\mathbb{R}^L_+$. Suppose that the indirect utility is also differentiable. Then

$$x(p,w) = -\frac{\nabla_p v(p,w)}{\nabla_w v(p,w)} \quad or\ x_i(p,w) = -\frac{\partial v(p,w)/\partial p_i}{\partial v(p,w)/\partial w}$$

Proof. Differentiate $v(p, e(p,\bar{u})) = \bar{u}$ w.r.t p.
other proofs are presented in the book.

Roy's identity

We already had $\partial v/\partial w = \lambda$ and $\partial u/\partial x_i = \lambda p_i$. Thus, we can rewrite Roy's identity as

$$-\frac{\partial v}{\partial p_i} = \lambda x_i = \frac{x_i}{p_i}\frac{\partial u}{\partial x_i}$$

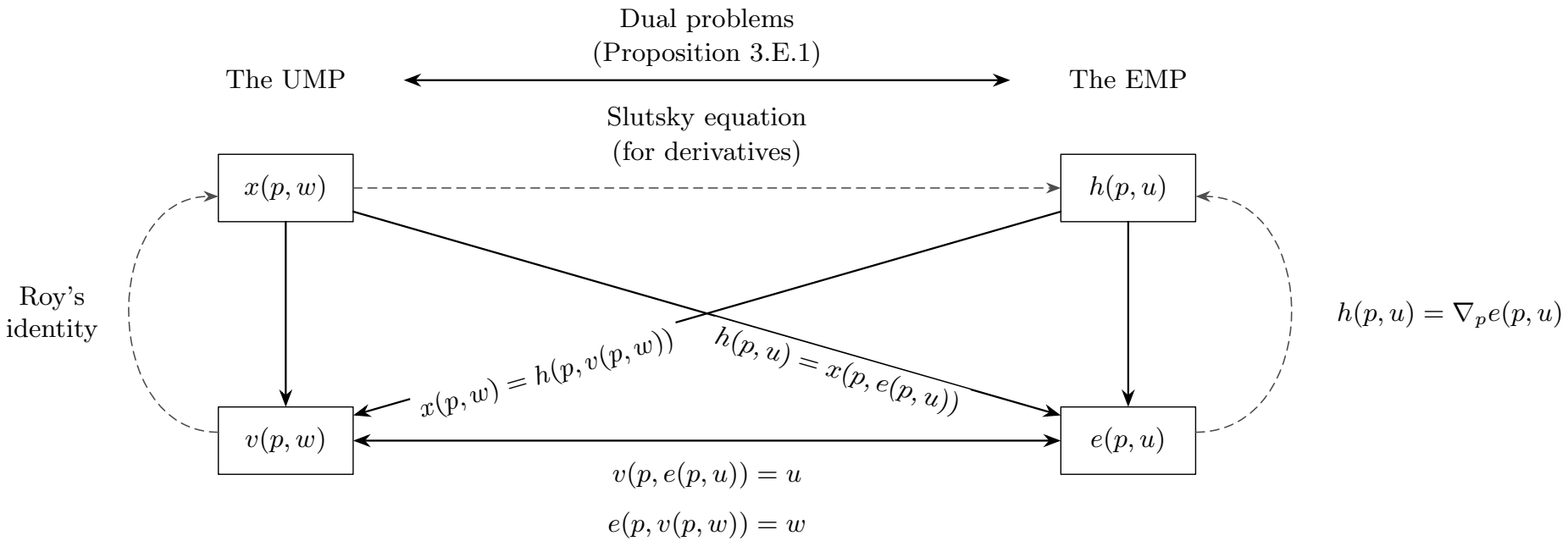
If the price of good i decreases by 1 dollar, the consumer has x_i dollars extra income.

If he spend this on good i, he can buy x_i/p_i more, and increase his utility by $(x_i/p_i) \times (\partial u/\partial x_i)$.

In general, the consumer would spend the extra income on a variety of goods, and substitute good 1 with others.

Roy's identity says such substitution is not important for measuring change in utility, and for that matter, we can assume spending all extra income on the good i.

Relationships between EMP and UMP



Integrability

If we have a demand function that is homogeneous of degree zero, satisfies Walras' law, and its $S(p, w)$ is symmetric and negative semi-definite, can we find a preference relation that rationalizes $x(p, w)$? Yes!

Indeed, these conditions are necessary and sufficient.

Is satisfying weak axiom sufficient for assuring the existence of rational preference?

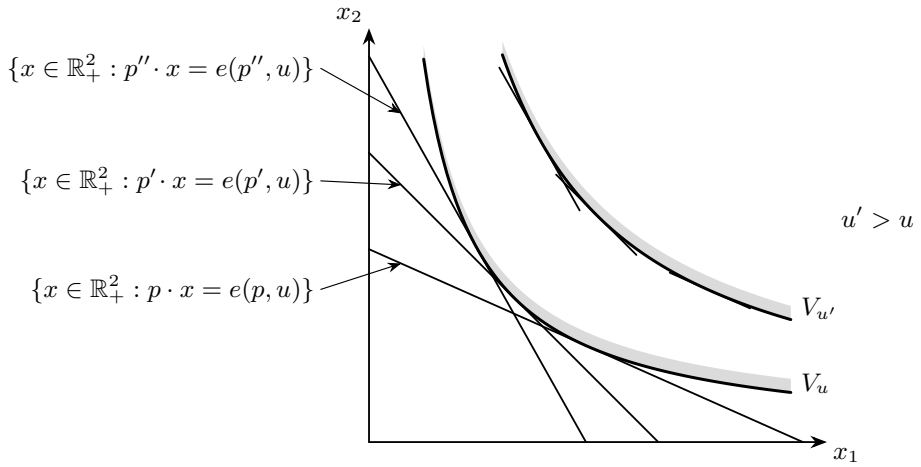
Why do we care?

- For measuring welfare effects we need to know consumer preference and choice is not enough
- Instead of deriving demand function from a utility that represent the preference, we can directly begin by a demand function and check the necessary and sufficient conditions.

Recovering $\succsim$ from $e(p,u)$ and $e(p,u)$ from $x(p,w)$

Change the EMP to $e(p,u) = \min_x \ p.x \ \text{ s.t. } \ x \in V_u$

Proposition 3.H.1 shows that V_u is the at-least-as-good-as sets that define a preference relation $\succsim$ with $e(p,u)$ as its expenditure function.



To recover $e(p,u)$ from $x(p,w)$, we need to solve the system of differential equations generated by $D_p e(p) = x(p, e(p))$.

It has a solution if $D_p^2 e(p) = S(p,w)$ is symmetric.

Recovering $u(x)$ from $v(p, w)$

Proposition

Suppose that $u(x)$ is quasiconcave and differentiable with strictly positive partial derivatives. Then $u(x) = \min_p v(p, p.x)$

Since $v(p, w)$ is homogeneous from degree zero, we can write $v(p, w) = v(p/w, 1) = v(p', 1)$ where $p' = p/w$.

Then we can solve $u(x) = \min_p v(p', 1)$ s.t. $p'.x = 1$ to find the utility function.

Welfare evaluation

Measuring the changes in consumer's well-being

For this we need to know preferences and cannot proceed with only weak axiom.

From now on, we assume a rational, continuous, locally nonsatiated preference and differentiable $u(.)$ and $v(.,.)$.

We focus on a welfare effect of a price change.

$$w: \text{ constant}, \quad p^0: \text{ initial price}, p^1: \text{ new price}$$

How can we evaluate the welfare effect of this change?

Welfare evaluation

If we know $\succsim$, we can build $v(p, w)$ and the sign of $v(p^1, w) - v(p^0, w)$ gives the direction of welfare change.

What about the level of change?

Welfare evaluation

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What about the level of change?

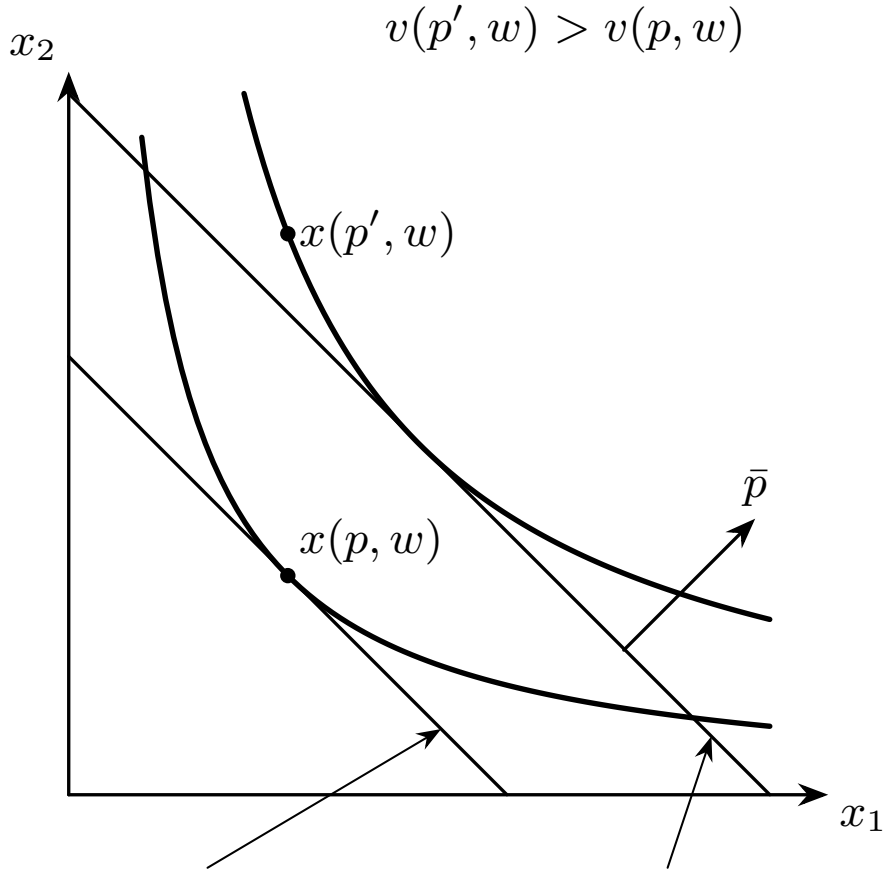
We can choose any $v(p, w)$, but a good choice is a function with monetary unit.

Then we can interpret the welfare effect in dollars.

Money metric indirect utility function

For an arbitrary numeraire price vector $\bar{p} >> 0$, $e(\bar{p}, v(p, w))$ is an indirect utility function in dollar units.

Then, we can measure welfare effect by $e(\bar{p}, v(p^1, w)) - e(\bar{p}, v(p^0, w))$.



Two measures for welfare change

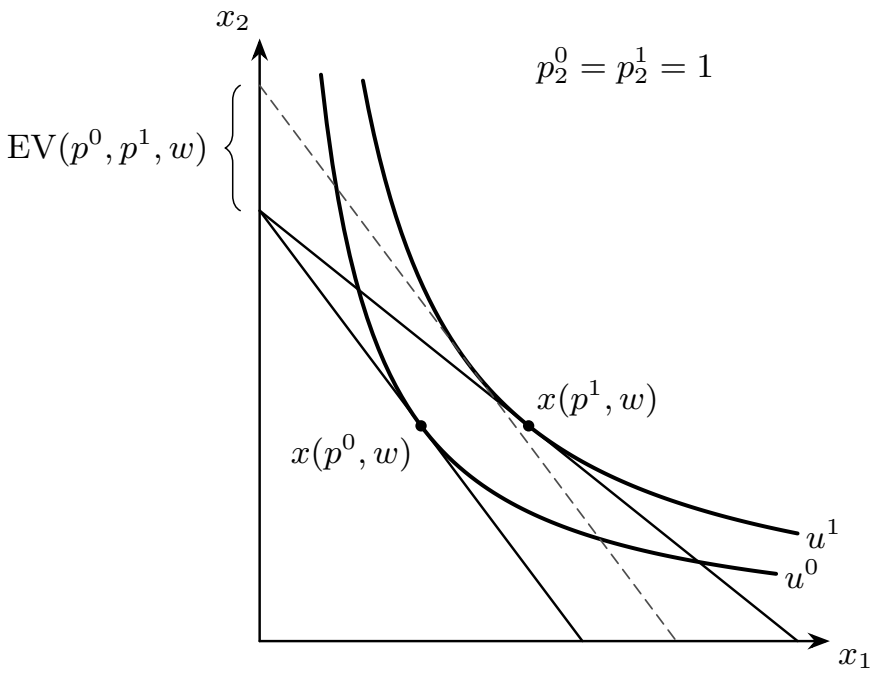
What are the natural candidates for $\bar{p}$?

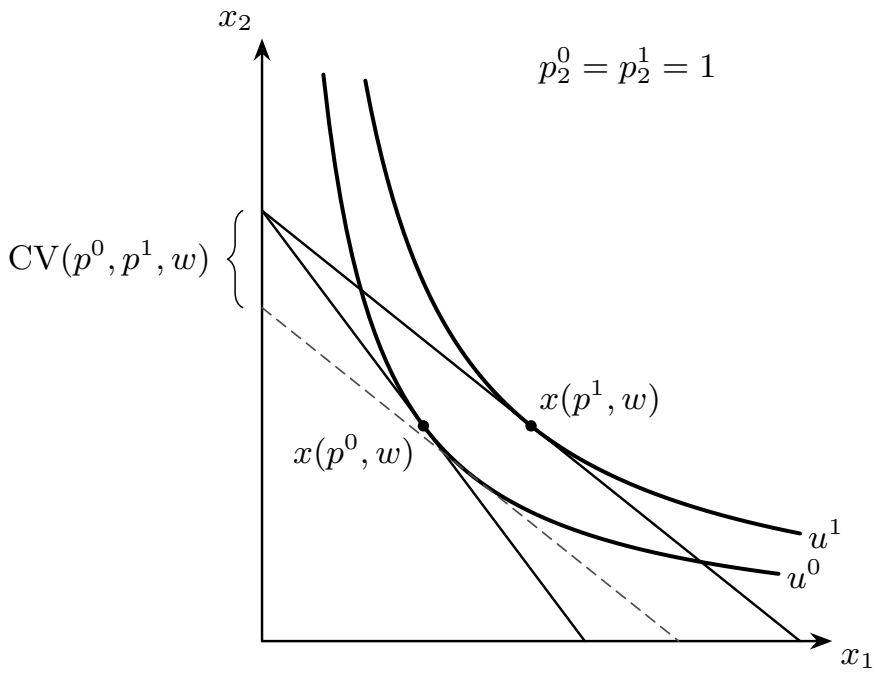
Two measures for welfare change

What are the natural candidates for $\bar{p}$? p^0 and p^1

equivalence variation: $EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0)$

compensating variation: $CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0)$





Equivalence vs. compensating variation

Both are welfare measures:
if $p^1 > p^0 \rightarrow EV < 0, CV < 0$ and if $p^1 < p^0 \rightarrow EV > 0, CV > 0$

$$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0) = e(p^0, u^1) - w = e(p^0, u^1) - e(p^1, u^1)$$

EV says how much are you willing to pay/accept to remain at p^0 ?

$p^1 > p^0$: how much to pay to prevent the loss of the price increase?
$p^1 < p^0$: how much to accept to forgo the gain of the price decrease?

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0) = w - e(p^1, u^0) = e(p^0, u^0) - e(p^1, u^0)$$

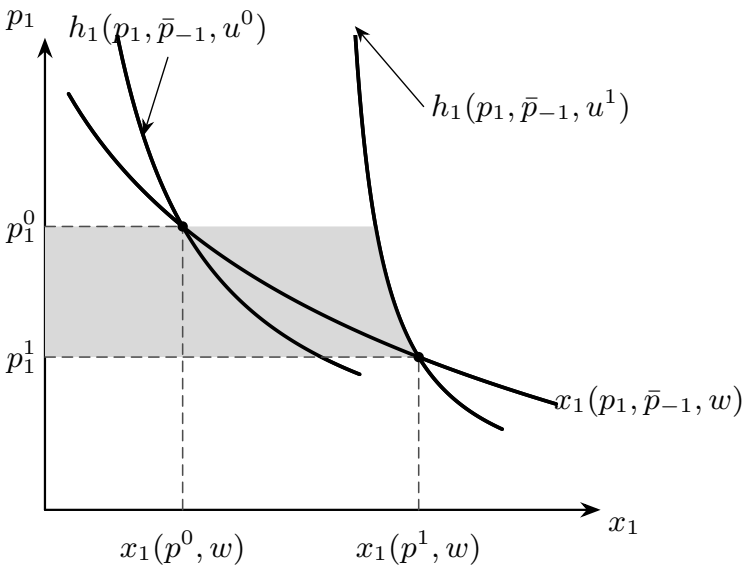
CV says how much are you willing to pay/accept to stay at p^1 ?

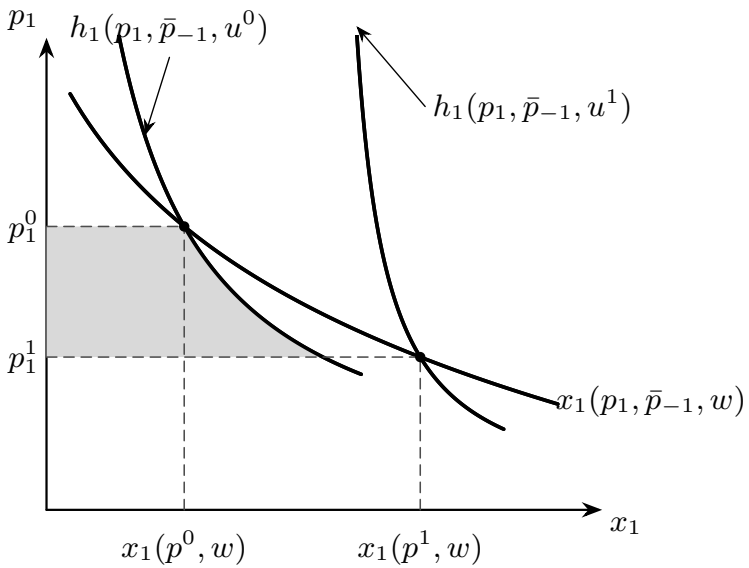
$p^1 > p^0$: how much to accept to tolerate the loss of the price increase?
$p^1 < p^0$: how much to pay to secure the gain of the price decrease?

Equivalence vs. compensating variation

If only price of good 1 changes:

$$EV = \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^1) dp_1 \qquad CV = \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^0) dp_1$$





DWL of a commodity tax

A new tax on good 1: $p_1^1 = p_1^0 + t$, makes revenue $T = tx_1(p^1, w)$.

An alternative is a lump-sum tax equal to T.

The welfare change

- from commodity tax $= EV(p^0, p^1, w) = e(p^0, u^1) - w$
- from lump-sum tax $= -T$

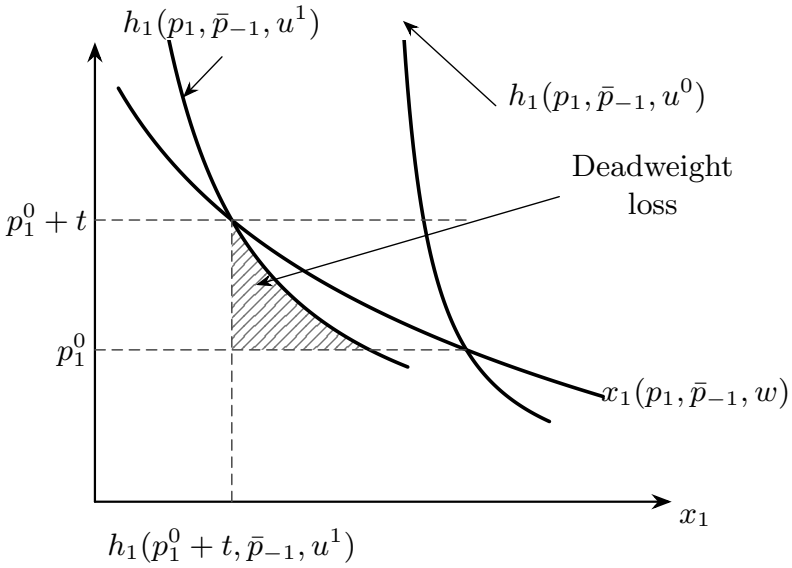
If $EV < -T$, the consumer is worse off by the commodity tax.

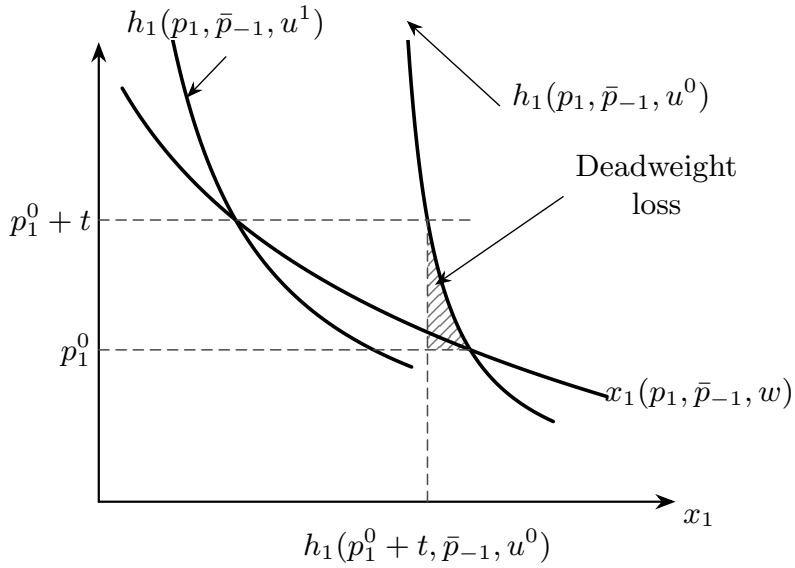
Dead-weight loss (DWL) is the extra amount by which the consumer is worse off. $DWL^1 = -T - EV = w - T - e(p^0, u^1)$

$$DWL^1 = \int_{p_1^0}^{p_1^0+t} [h_1(p_1, \bar{p}_{-1}, u^1) - h_1(p_1^0 + t, \bar{p}_{-1}, u^1)]dp_1$$

In a similar way we can measure DWL^0 using $-CV - T$ for u^0.

DWL in a figure





Using EV vs. CV

When we compare two prices that one is the numeraire, both are fine:

$$\text{if } p^1 > p^0 \rightarrow EV(p^0, p^1, w) < 0, CV(p^0, p^1, w) < 0, \text{ and vice-versa.}$$

When we compare two prices that none of them is the numeraire:

$$EV(p^0, p^1, w) - EV(p^0, p^2, w) = e(p^0, u^1) - e(p^0, u^2)$$

$$CV(p^0, p^1, w) - CV(p^0, p^2, w) = e(p^2, u^0) - e(p^1, u^0)$$

Using EV vs. CV

When we compare two prices that one is the numeraire, both are fine:

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When we compare two prices that none of them is the numeraire:

$$EV(p^0, p^1, w) - EV(p^0, p^2, w) = e(p^0, u^1) - e(p^0, u^2)$$

$$CV(p^0, p^1, w) - CV(p^0, p^2, w) = e(p^2, u^0) - e(p^1, u^0)$$

To have a same indirect utility, we must have a fixed price as numeraire.

$\Rightarrow$ just EV can be used.

Welfare evaluation with limited information

To measure welfare impact when we have demand:

$$x(p,w) \xrightarrow{\text{integrability}} e(p,u) \Rightarrow EV \text{ and } CV$$

When information is limited: we only know p^0, p^1, and $x^0 = x(p^0, w)$.

Proposition

Suppose that the consumer has a locally nonsatiated rational preference. If $(p^1 - p^0)x^0 < 0$, the consumer is strictly better off under (p^1, w).

Proof: $(p^1 - p^0)x^0 < 0 \rightarrow p^1 x^0 < w \rightarrow x^0$ is affordable under p^1 and not on the border of budget set. By local nonsatiation, there is a better consumption bundle under p^1.

Approximating welfare measures

Hicksian demand is not observable and for welfare we need to solve differential equations to find $e(p, u)$.

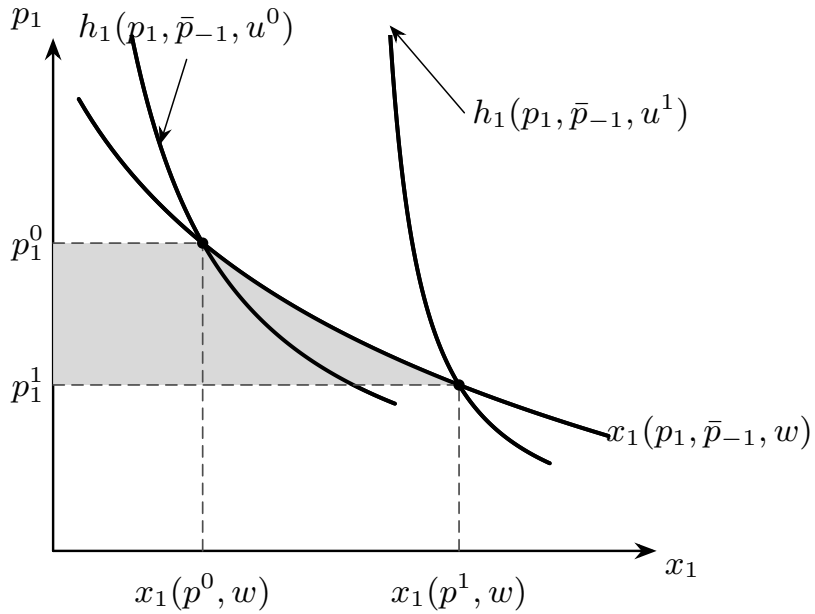
Can we approximate the welfare using integrals of $x(p, w)$ instead of $h(p, u)$?

Average variation:

$$AV(p^0, p^1, w) = \int_{p_1^1}^{p_0^1} x(p_1, \bar{p}_{-1}, w) dp_1$$

$CV \leq AV \leq EV$ for normal goods.

If no wealth effect: CV=AV=EV
when that happens?



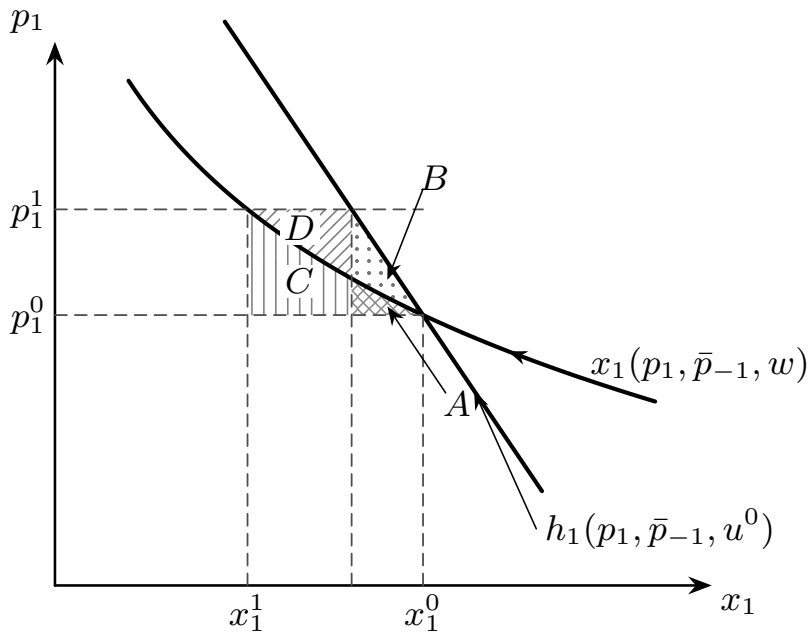
If wealth effect or Δp are small, still AV gives a good approximation.

Average variation approximation

The error can be quite large if measured as fraction of DWL:

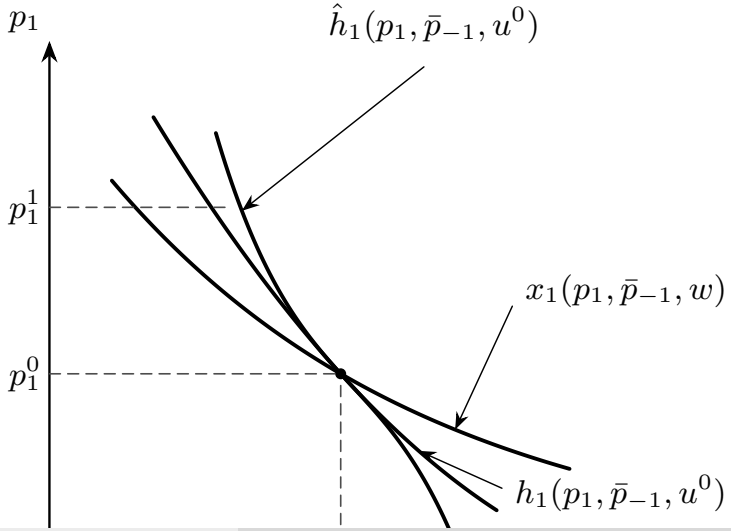
The approximation $= A + C$

The real amount $= A + B$



Better approximation for DWL using Taylor expansion of $h(p, u^0)$ at p^0:

$$\tilde{h}(p, u^0) = x(p^0, w) + S(p^0, w)(p - p^0)$$



The strong (Houthakker) axiom of revealed preferences

Definition

The market demand function $x(p, w)$ satisfies the strong axiom of revealed preferences (SARP) if for any list $(p^1, w^1), \ldots, (p^N, w^N)$ with $x(p^n, w^n) \neq x(p^{n+1}, w^{n+1})$ for all $n < N$, we have $p^N.x(p^1, w^1) > w^N$ whenever $p^n.x(p^{n+1}, w^{n+1}) \leq w^n$ for all $n < N$.

Proposition

If $x(p, w)$ satisfies the SARP, then there is a rational preference relation that rationalizes $x(p, w)$, such that
$\forall y \neq x(p, w)$ *and* $y \in B_{p,w}$, $x(p, w) \succ y$.

Therefore, the choice-based theory founded on the SARP is essentially equivalent to the preference-based theory of demand.